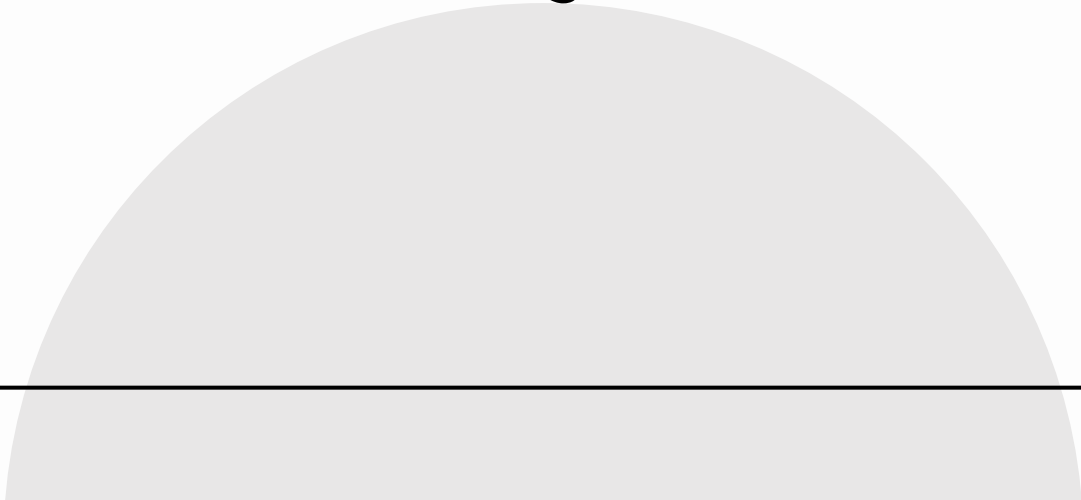


# **Data Science**

## **Effective Data Visualization**

Prof. Rayson Laroca  
rayson@ppgia.pucpr.br

# **Exploratory vs. Explanatory Analysis**



# Exploratory vs. Explanatory

- Exploratory analysis
  - Used to understand the data
  - Questions emerge during the analysis
  - Statistics and visualizations help us inspect patterns
  - The output is often rough and provisional
- Explanatory analysis
  - Used to communicate insights clearly
  - Refines what was learned during exploration
  - Emphasizes the main message
  - Usually supports a story, decision, or recommendation

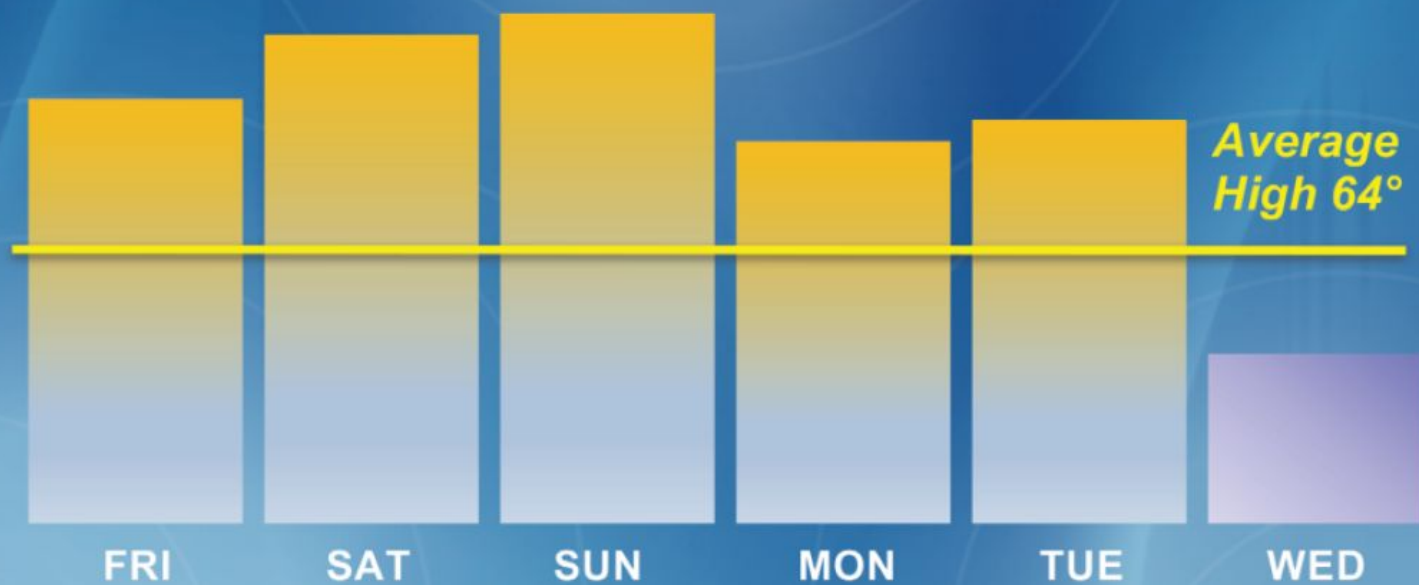
# Where visualization fits in the data workflow

- Data extraction
- Data cleaning
- Exploratory analysis
- Data analysis
- **Communication and sharing**
  - Once you have results, how do you present them **effectively**?

Look at the next chart for 1-2 minutes.

## WEEKLY FORECAST

WEATHER TEAM 4



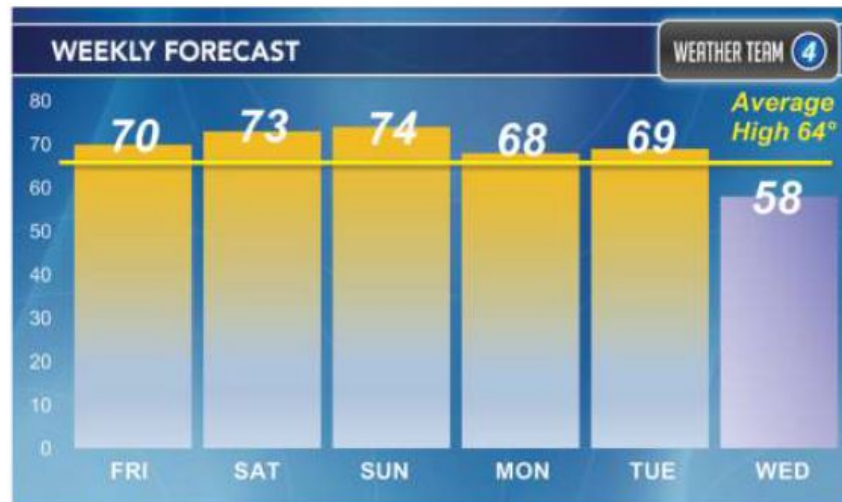
Try to estimate  
Sunday's high temperature.

## WEEKLY FORECAST

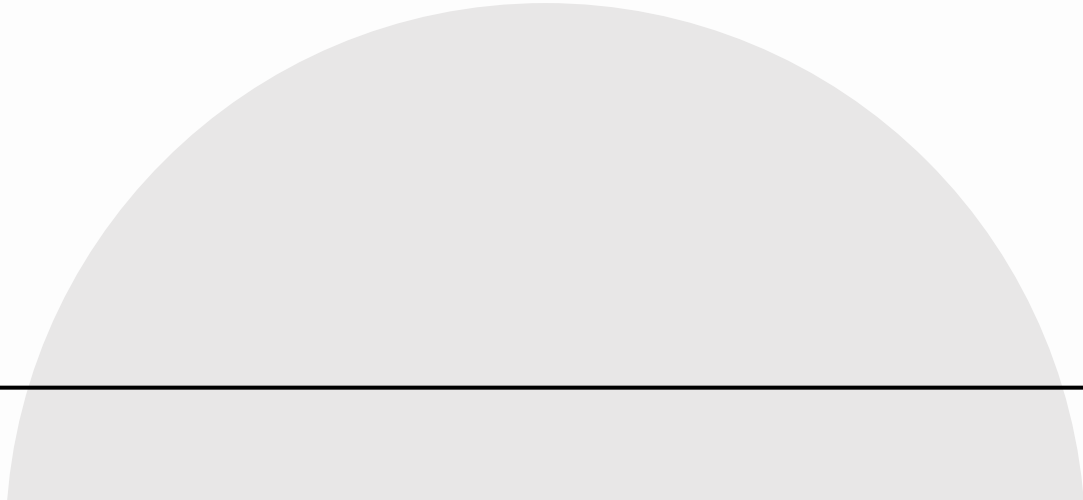
WEATHER TEAM

4





# **EFFECTIVE DATA VISUALIZATION**



# Effective Data Visualization

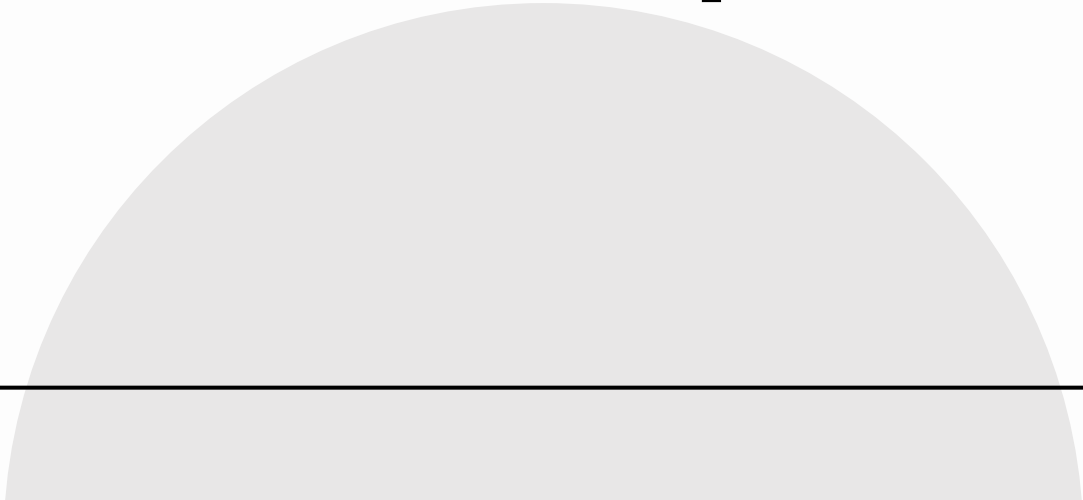
- Visualizations are communication tools
  - Therefore, the reader should understand the message we intend to communicate
- A useful habit is to check every chart before presenting it
- We will learn this by analyzing what **NOT** to do

# A quick checklist for better charts

- Title
- Axis labels
- Units
- Legend
- Scale
- Ordering
- Colors
- Text size
- Chartjunk

See [Evergreen's checklist](#) on  
Canvas for more details.

**What makes a  
visualization poor?**

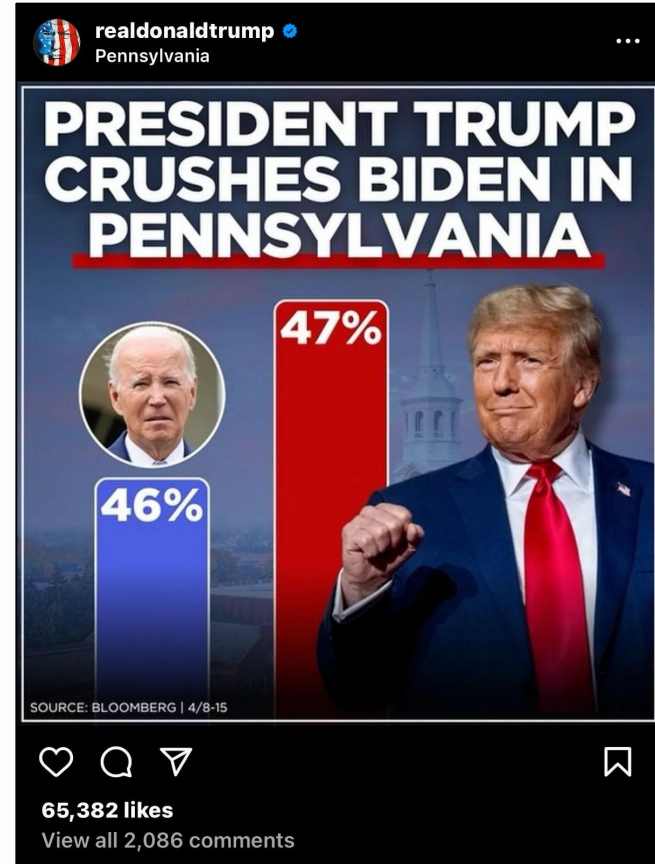


# What makes a visualization poor?

- A visualization is poor when its message is unclear
- Common problems:
  - Ambiguity
  - Missing information
  - Misleading emphasis
  - Unnecessary distraction

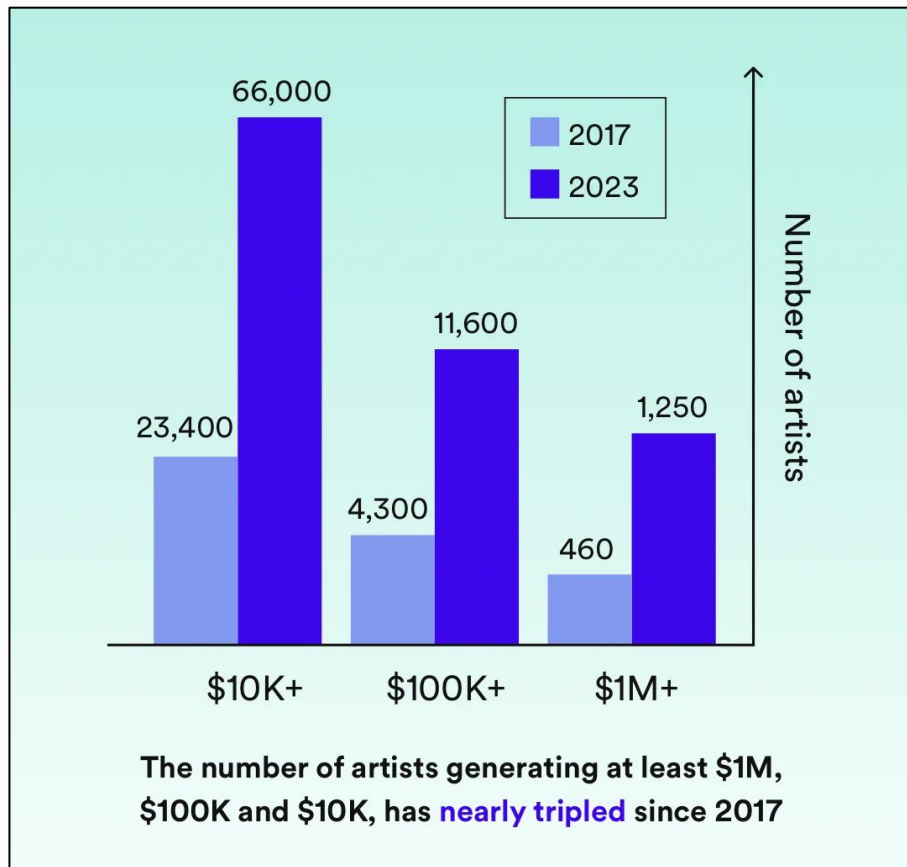
# Example

- Improper use of the **y-axis**
  - The difference between the values is relatively small, but the scale gives the impression that it's much larger than it actually is.
    - Misleading scales exaggerate differences



# Example

- Improper use of the **y-axis**
  - Who needs a consistent y-axis?  
Certainly not Spotify...
  - Inconsistent axes distort comparison

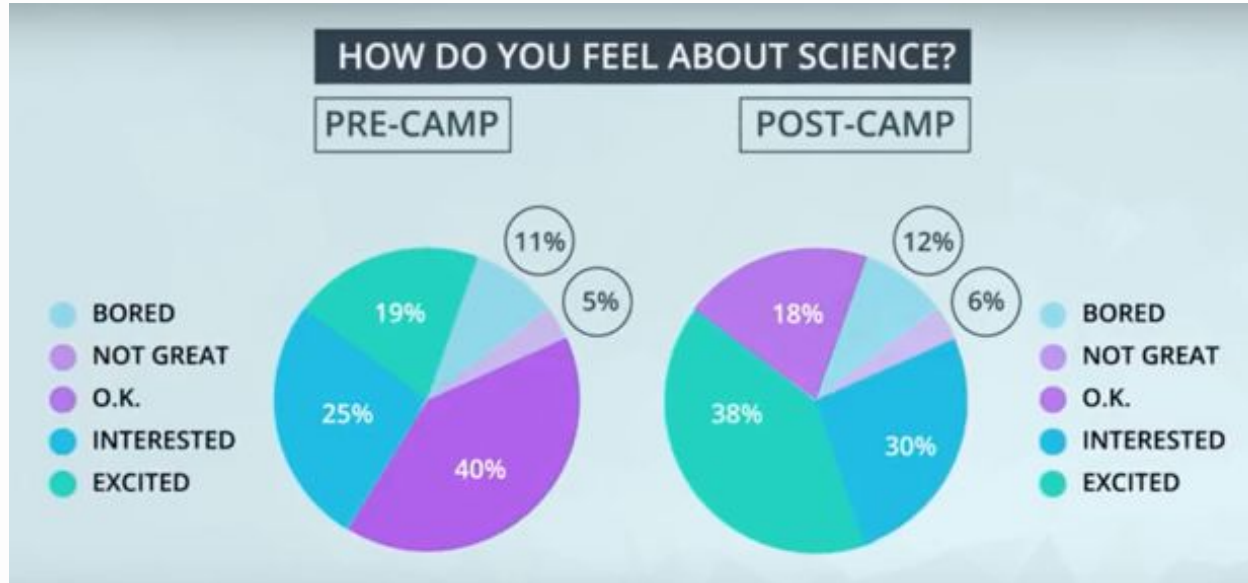


# Poor Visualization

- These examples are easy to spot because the distortion is obvious
  - In more complex charts, misleading choices are often harder to notice

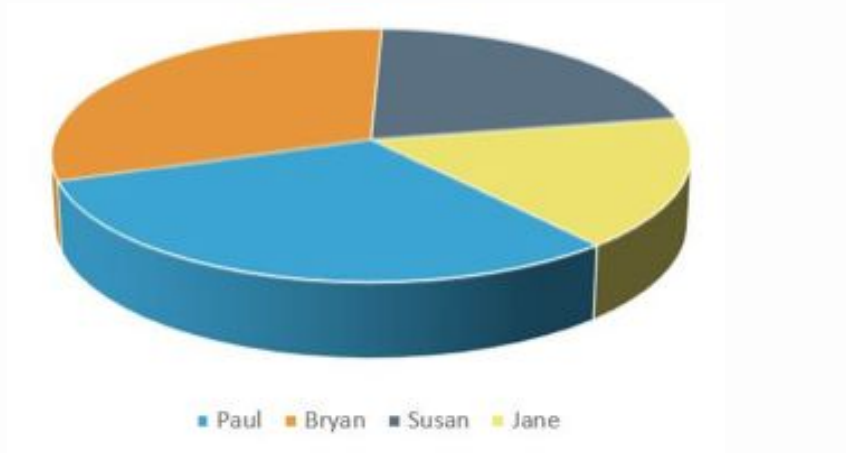
# Example

- Is it possible to say that interest in science increased as a result of the camp?

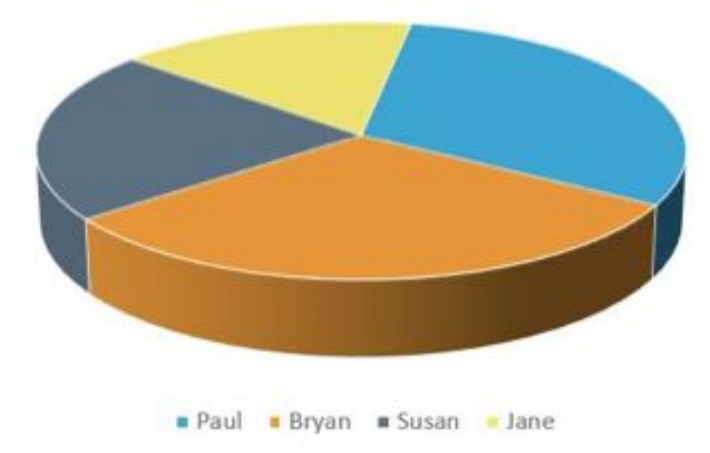


# Example

- Which of the regions below is larger?



Bryan or Paul?

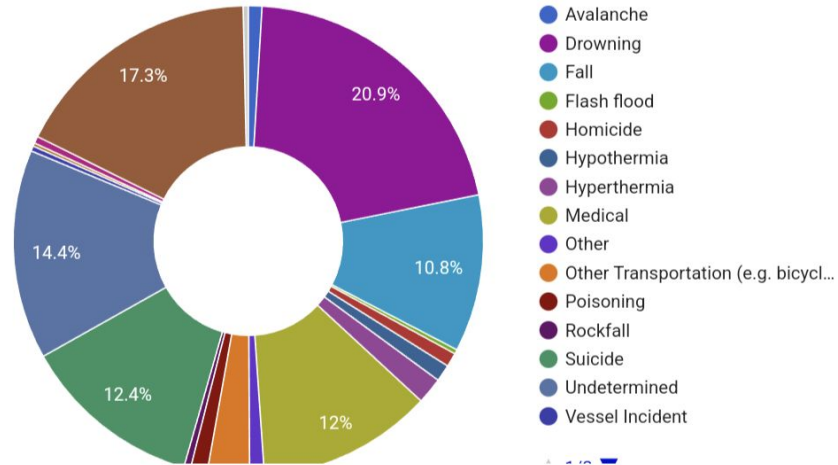


What about here?

# Example

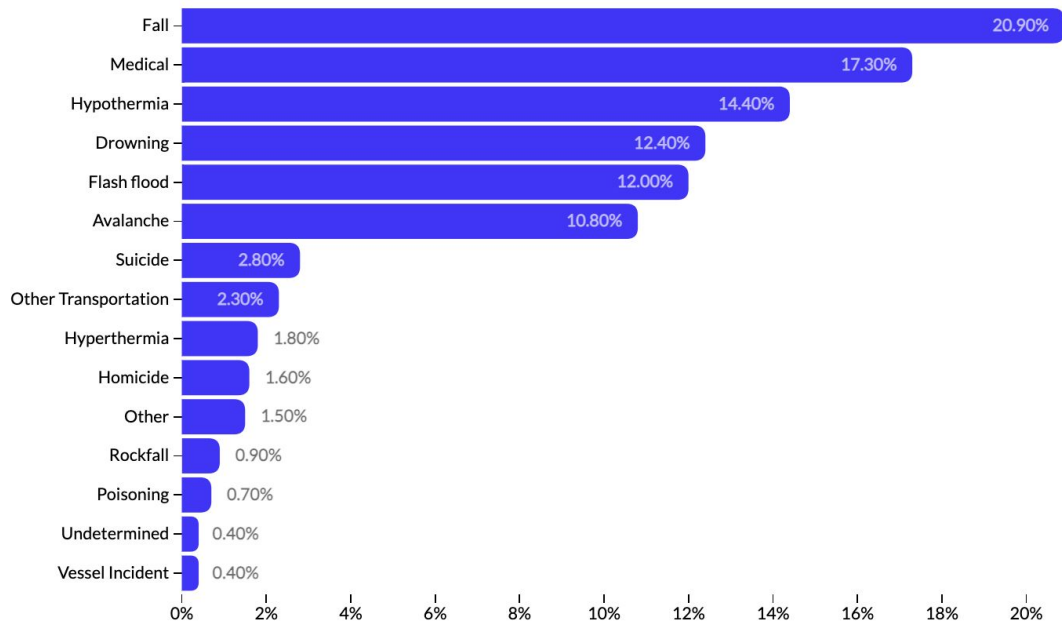
## What Are the Top Causes of Death in the National Parks?

Fatalities in National Parks (2007-2023) by Cause of Death



# Example

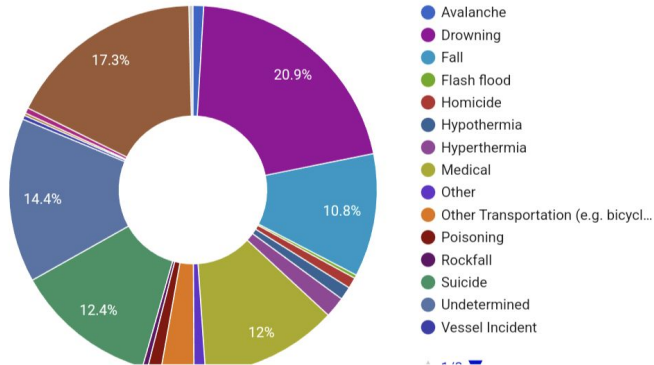
Top Causes Of Death In National Parks



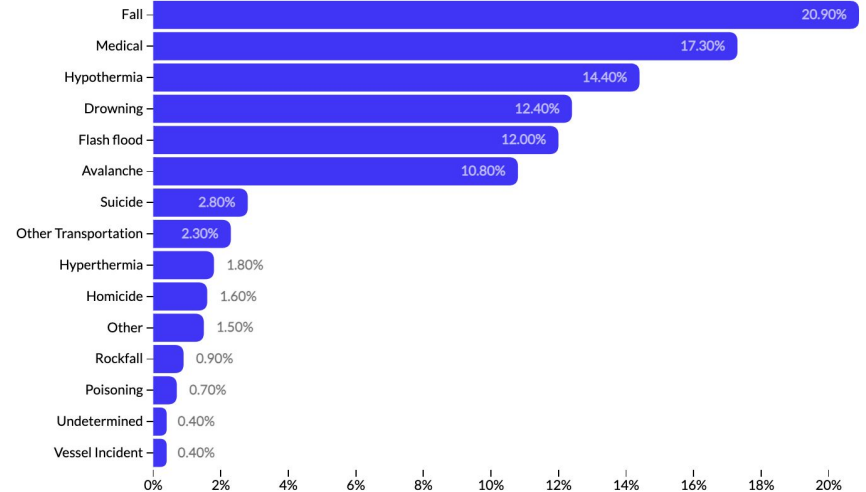
# Example

## What Are the Top Causes of Death in the National Parks?

Fatalities in National Parks (2007-2023) by Cause of Death

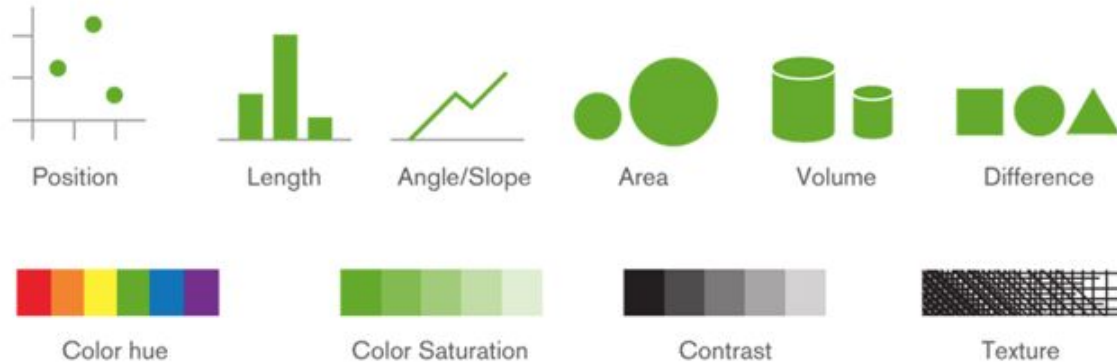


Top Causes Of Death In National Parks



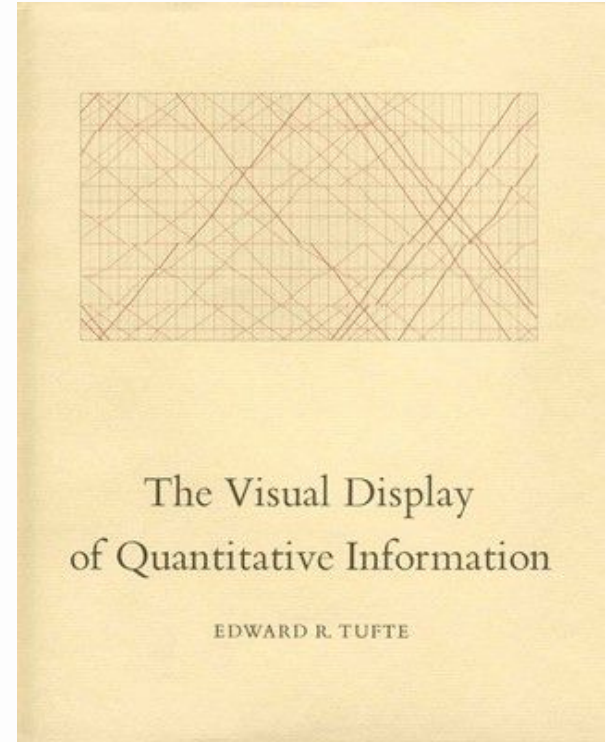
# Visual Components

- **Visual channels** used in data visualization:



# Chartjunk

- When designing a chart, it helps to think not only about what to include, but also about **what to remove**
- Tufte's data-ink ratio reminds us to maximize the ink that communicates data and minimize the ink that does not.



# Chartjunk

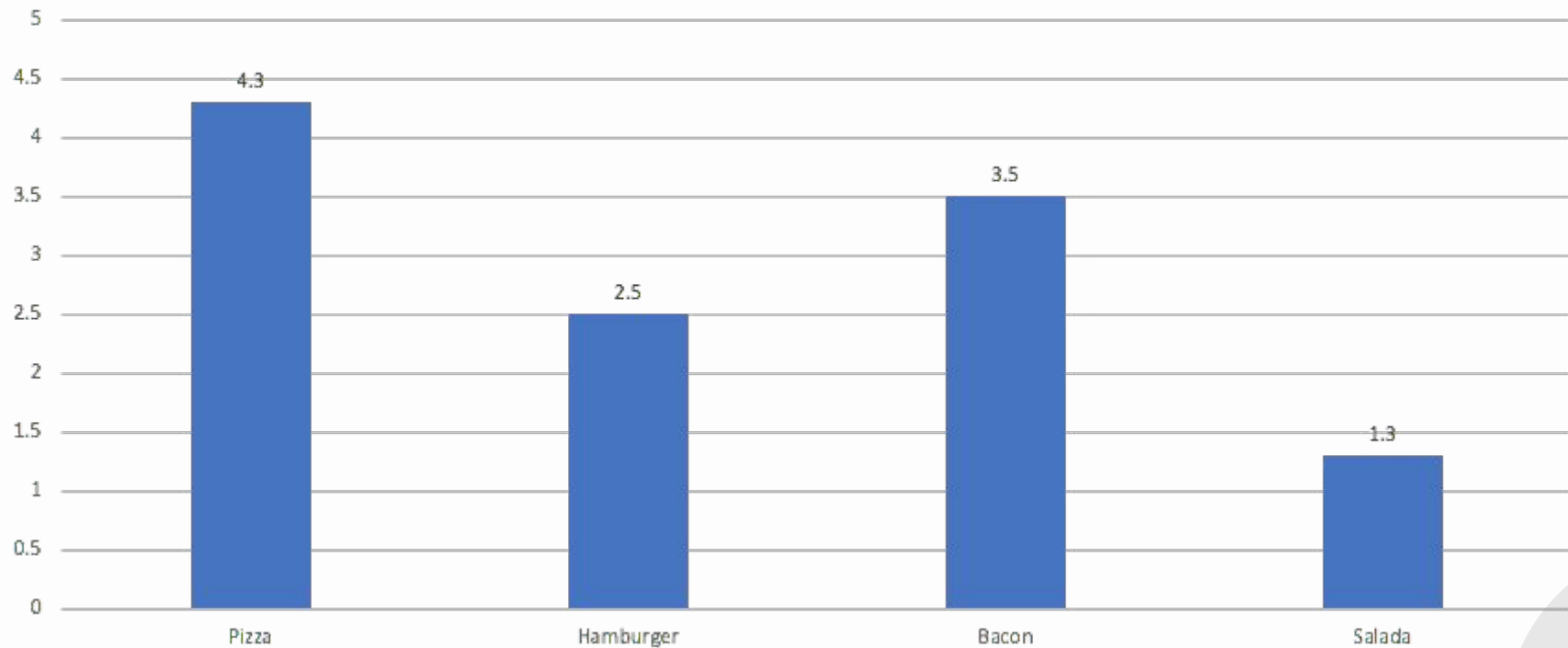
Too much decoration, too little clarity



# Chartjunk

A cleaner version

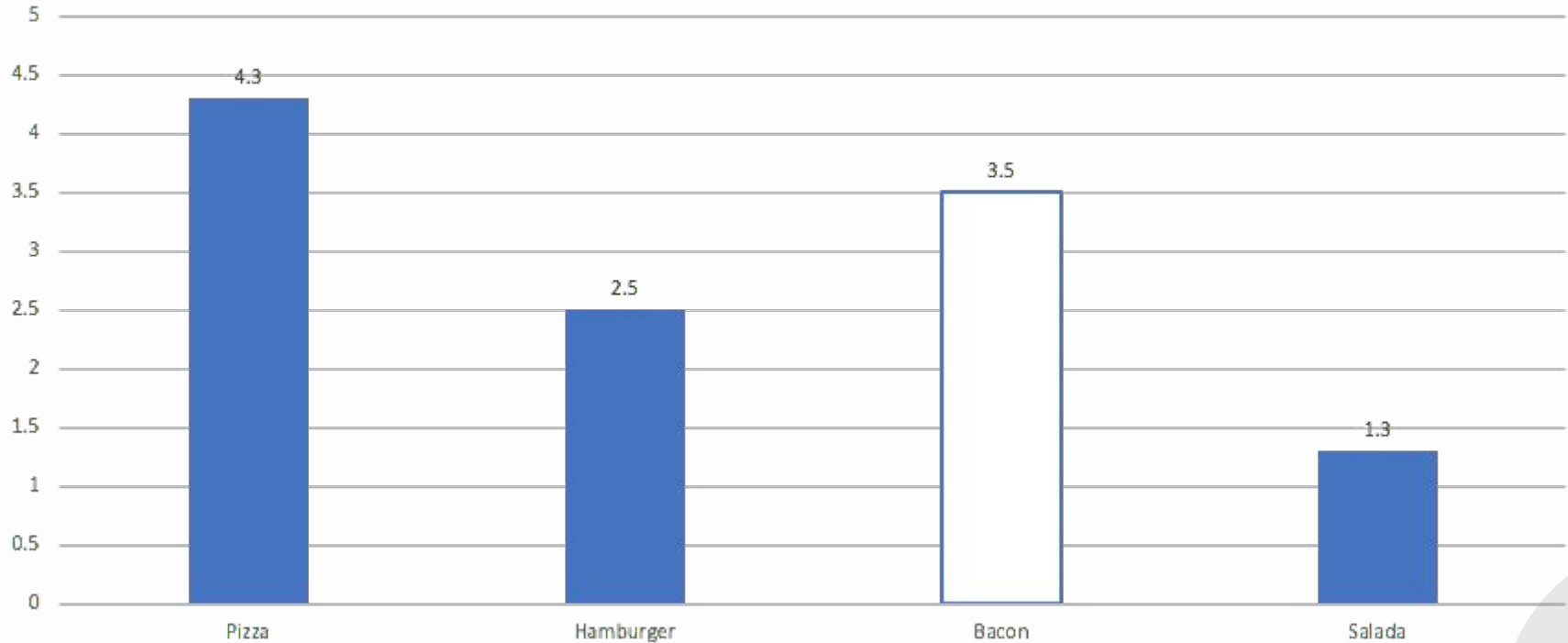
Preferência por alimento



# Chartjunk

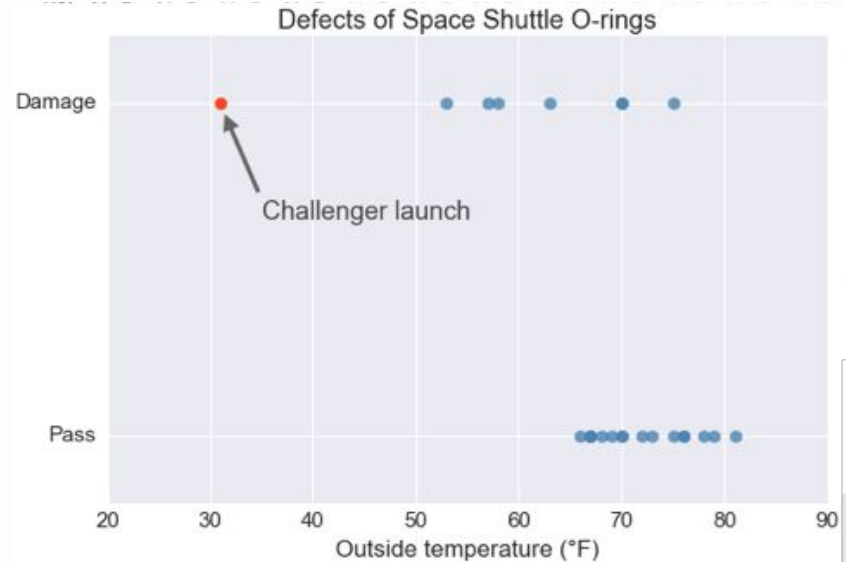
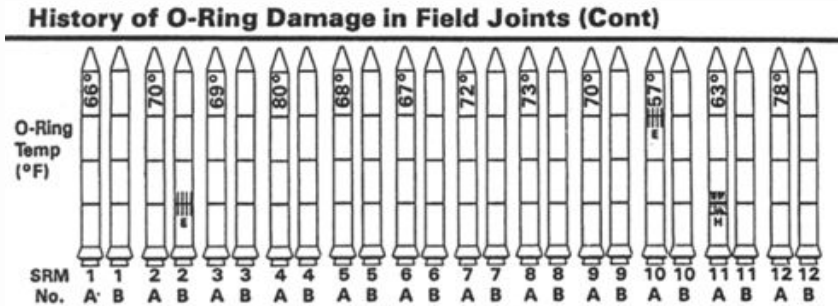
Use color to emphasize the category you want the audience to notice

Preferência por alimento



# Chartjunk

- Which plot helps you understand the pattern?

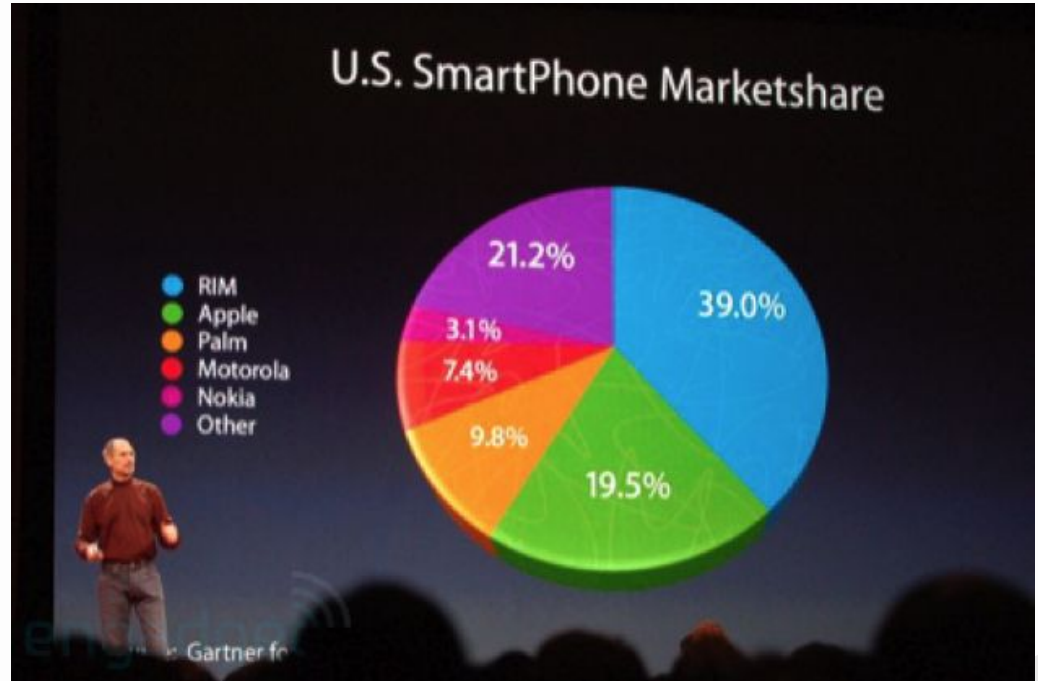


# **Practice: Critiquing Visualizations**



# What problems do you see here?

- A bar chart would support ranking better
- Color legend requires eye travel



# And here?

## THE WORLD CUP'S BIG GUNS

% OF TEAM'S RUNS SCORED BY TOP SCORER



# AMERICANS WHO HAVE TRIED MARIJUANA

CBS NEWS POLL

51%  
TODAY

43%  
LAST YEAR

34%  
1997



Source: MOE +/- 4%

**HIGH SUPPORT FOR LEGALIZING MARIJUANA**

MORE THAN HALF OF AMERICANS SAY THEY'VE TRIED POT



LIVE

**CBSN**

## Mortes de Covid-19 no Brasil

75% das vítimas tinham doenças associadas

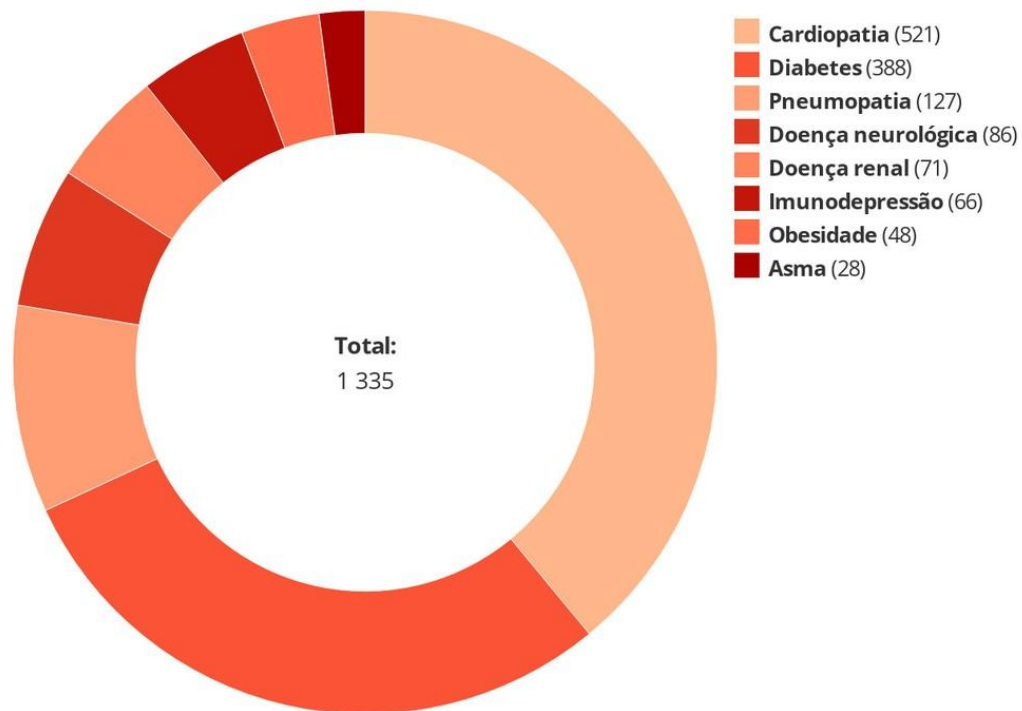


Gráfico: G1 • Fonte: Ministério da Saúde

## Mortes por coronavírus em pacientes com outras doenças

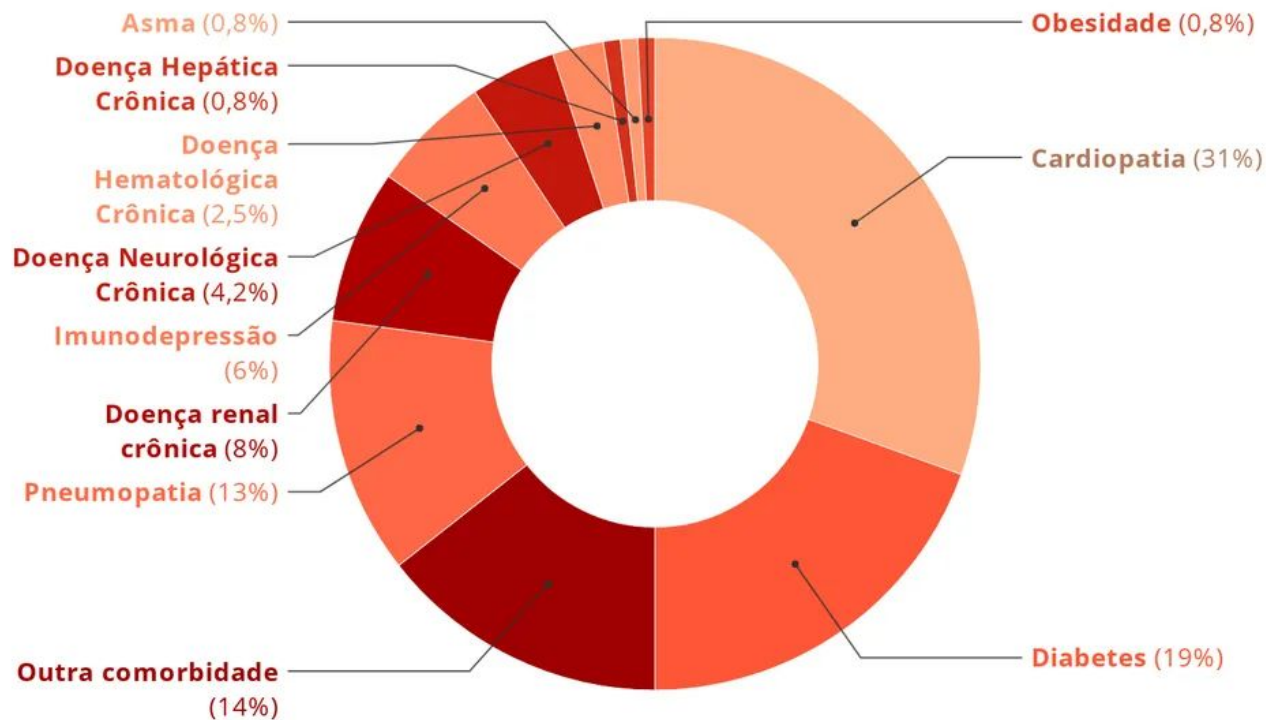
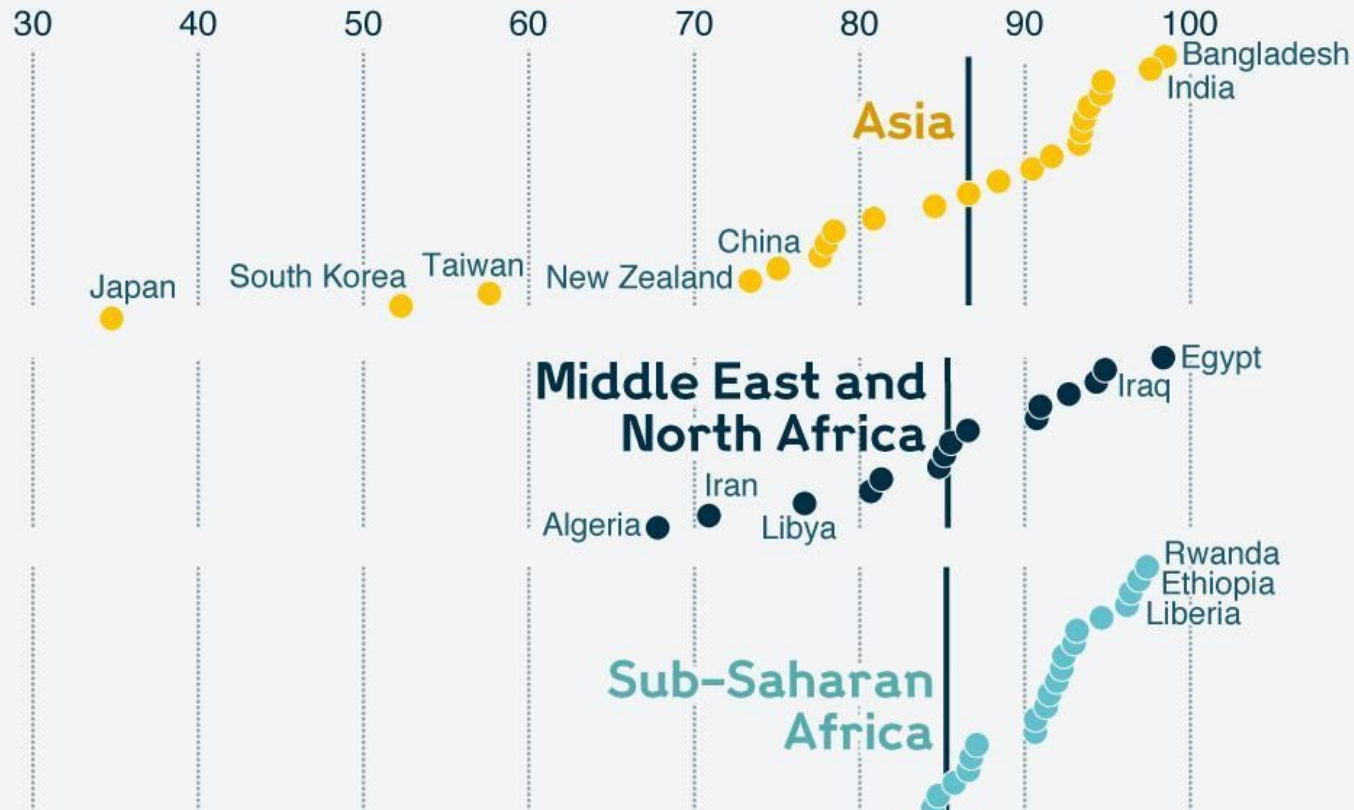


Gráfico: G1 • Fonte: Ministério da Saúde

# % of people who believe vaccines are safe, by country and global region

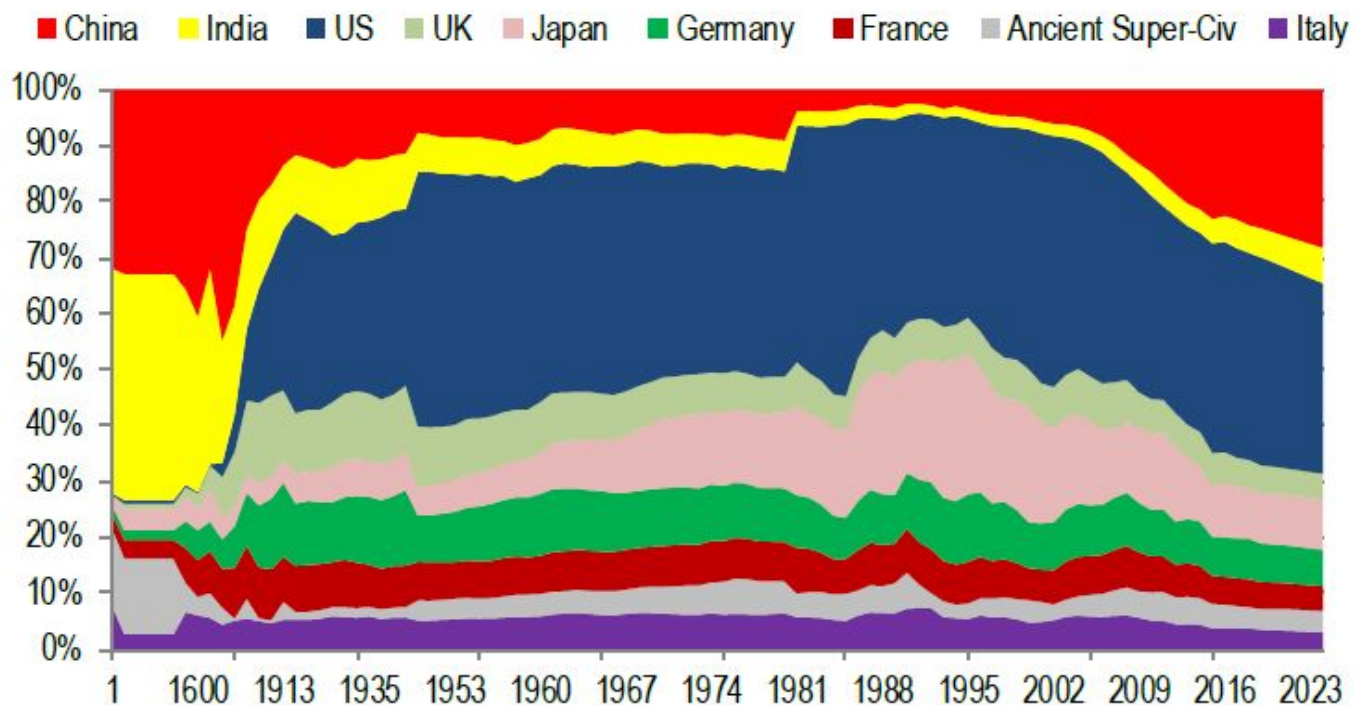
Dark vertical lines represent region medians



## Equal Number of Men & Women Voted In 2019



**The progression of world GDP – percentage of world GDP by major countries/regions. Are current trade tensions really an attempt to stymie China's impressive rise?**

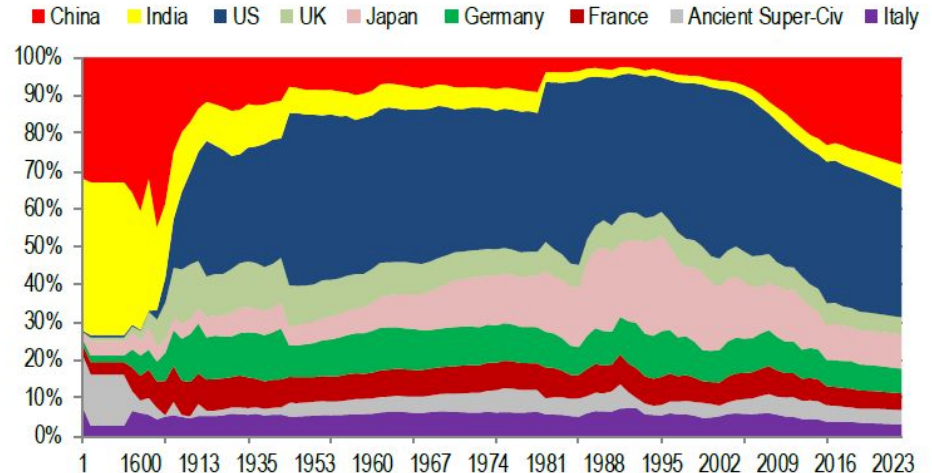


Source: BofA Merrill Lynch Global Research, Angus Maddison, IMF GDP data and estimates between 1980 and 2024. Ancient super-civilizations include Greece, Turkey, Algeria, Iraq, Egypt and Iran.

# What problems do you see here?

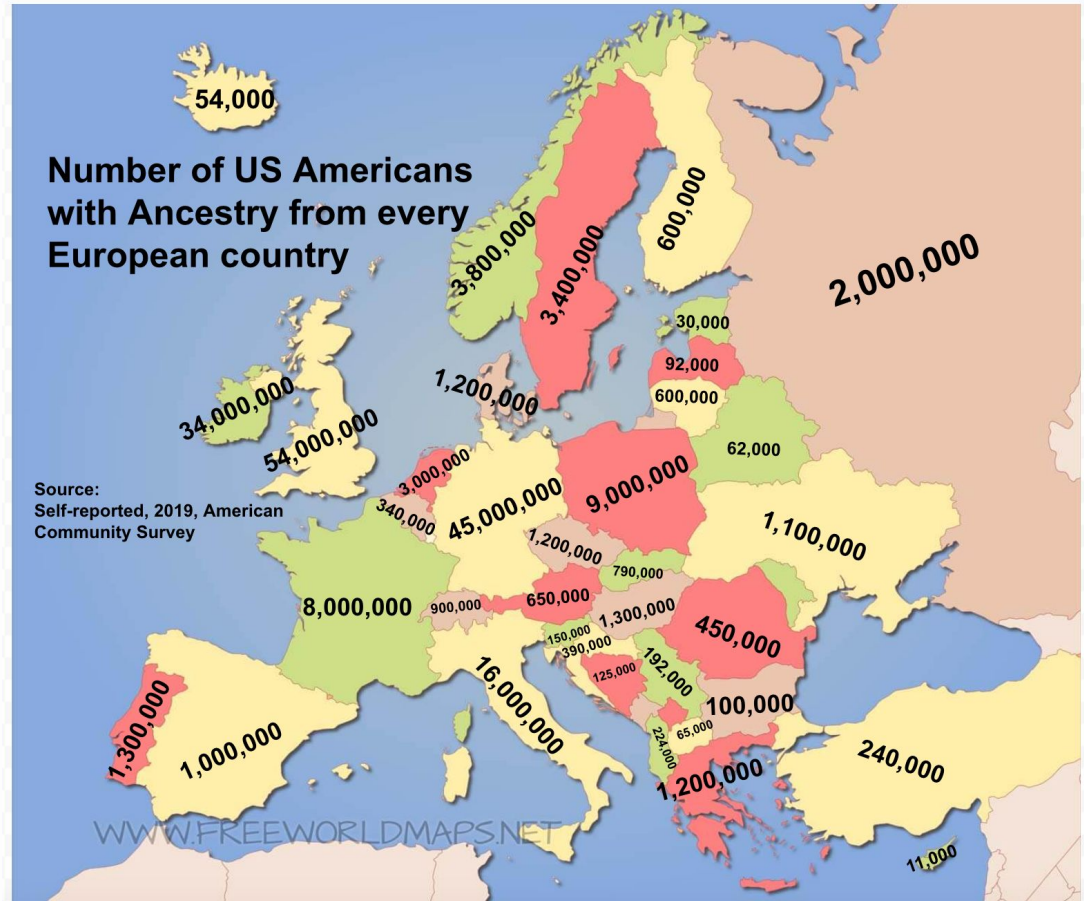
- The message is not immediately visible
  - Hard to track individual series over time
- Intense colors compete for attention

The progression of world GDP – percentage of world GDP by major countries/regions. Are current trade tensions really an attempt to stymie China's impressive rise?

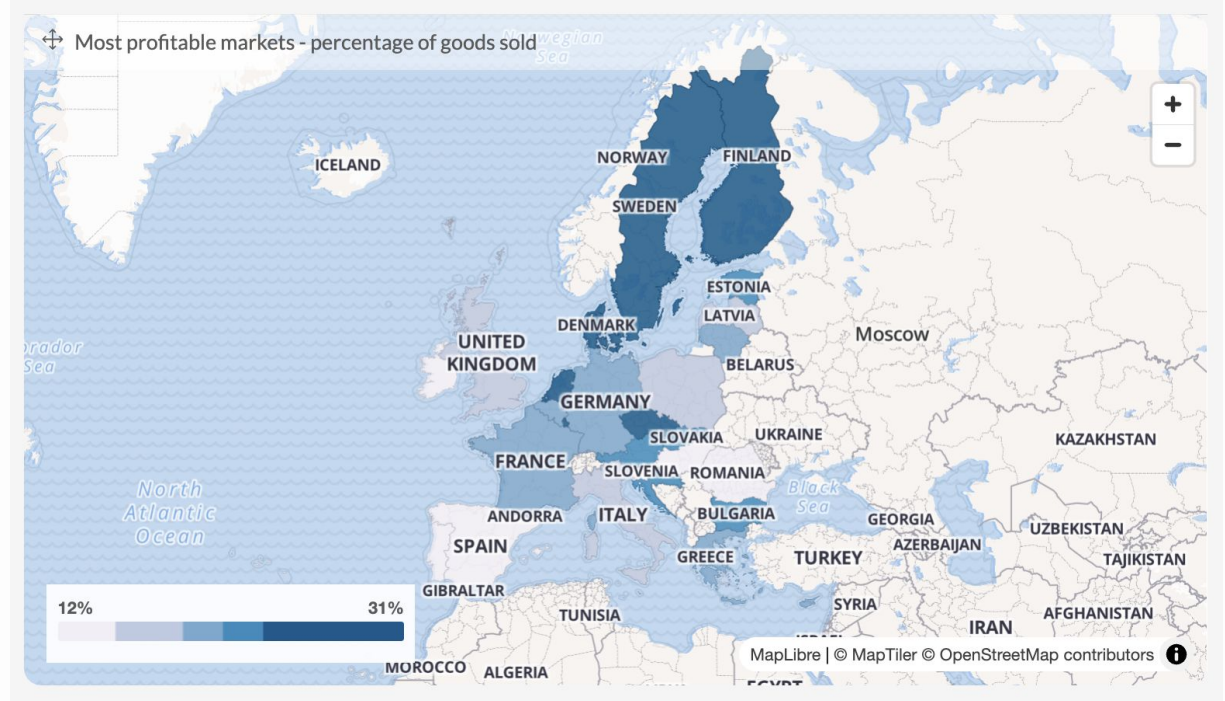


Source: BofA Merrill Lynch Global Research, Angus Maddison, IMF GDP data and estimates between 1980 and 2024. Ancient super-civilizations include Greece, Turkey, Algeria, Iraq, Egypt and Iran.

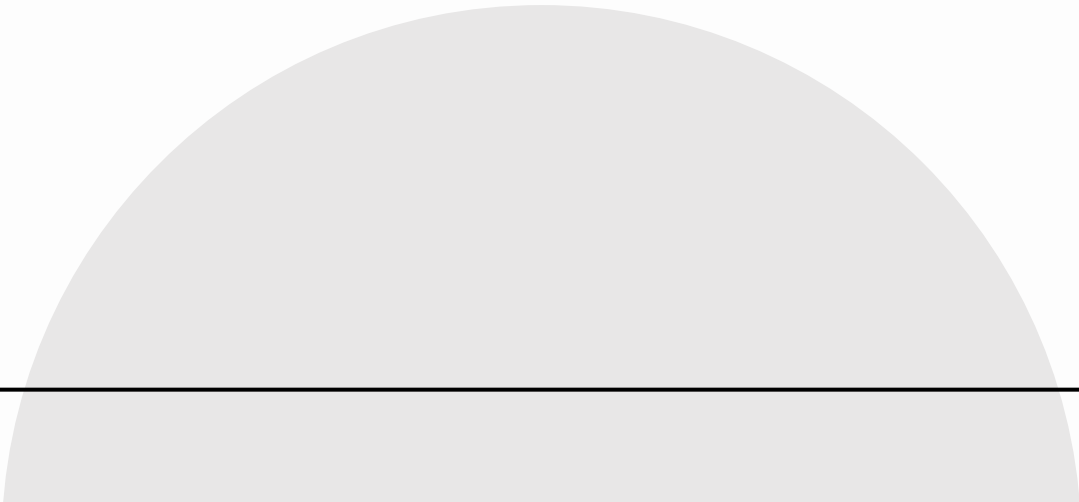
The colors make no sense as the same color is used for different values on the map.



Here is what a good  
use of colors looks like  
in a map chart:



**If you're not satisfied  
enough...**



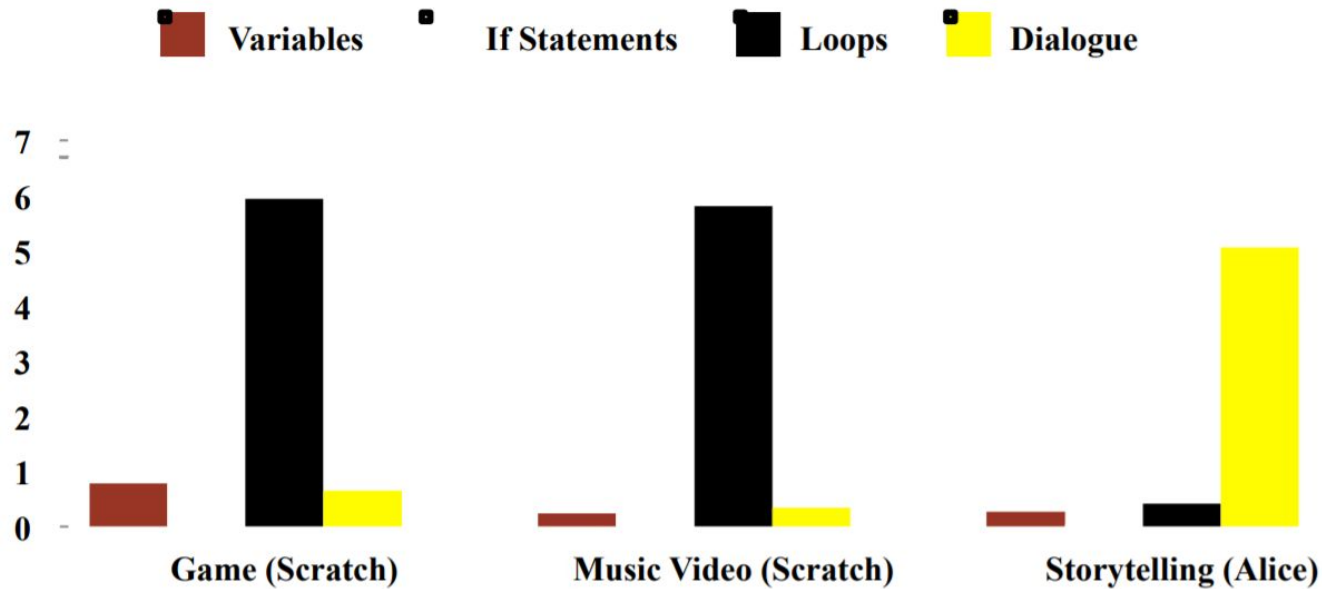
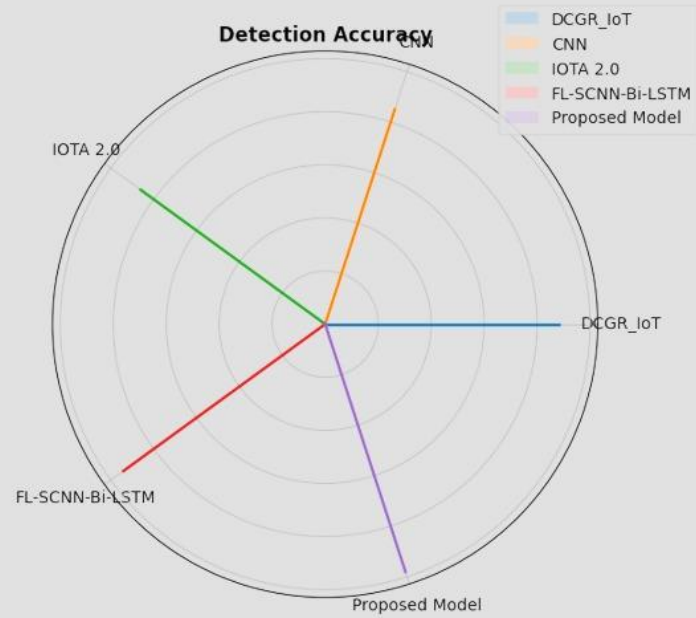


Figure 12. Use of Programming Constructs in Game, Music Video, and Storytelling Projects (Per 100 Lines of Code)

<https://dl.acm.org/doi/abs/10.1145/2157136.2157319>



**Figure 5. Detection Accuracy of the proposed model**

# References

- Most of these visualizations were obtained from
  - <https://badvisualisations.tumblr.com/>
  - <https://www.luzmo.com/blog/bad-data-visualization>

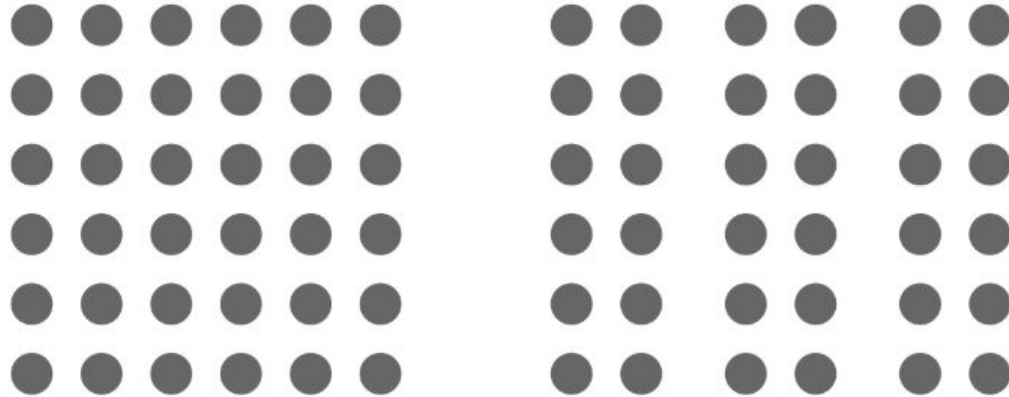
# **GESTALT PRINCIPLES**

# Gestalt Principles

- Proximity (*Proximidade*)
- Similarity (*Similaridade*)
- Enclosure (*Cerceamento*)
- Closure (*Fechamento*)
- Continuity (*Continuidade*)
- Connection (*Conexão*)

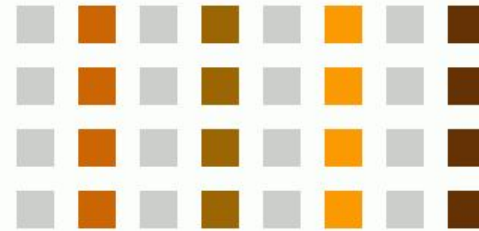
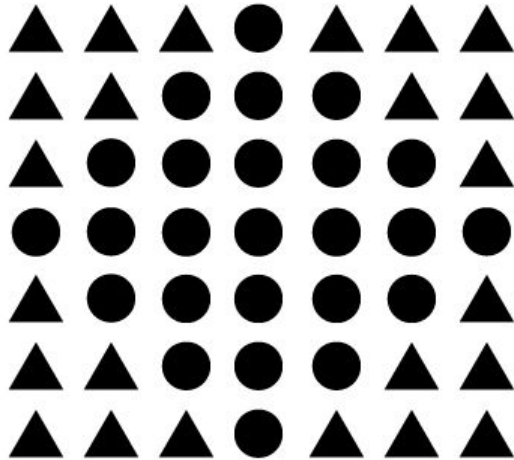
# Proximity (Proximidade)

- Things that are closer to one another are perceived as belonging to a same group



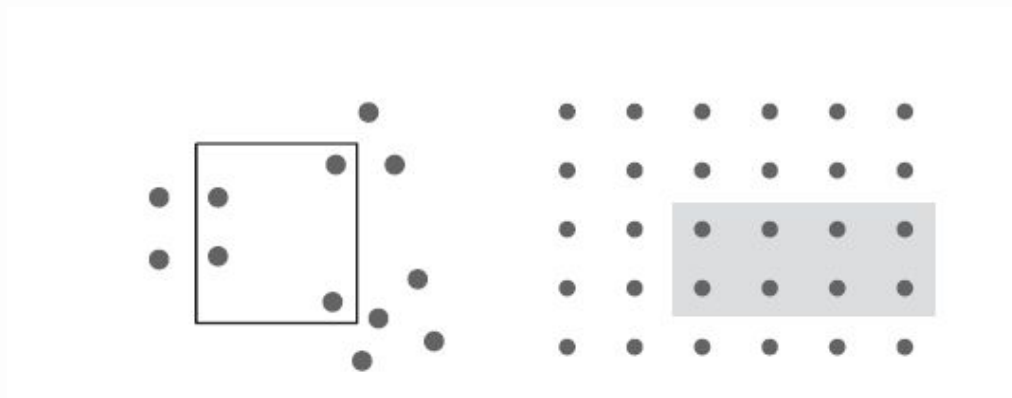
# Similarity (Similaridade)

- Objects that share shapes, sizes, colors or orientation are perceived as belonging to the same group



# Enclosure (Cerceamento)

- We observe objects that are enclosed together as belonging to the same group



# Closure (Fechamento)

- We perceive objects as a whole even though some parts are missing



# Continuity (Continuidade)

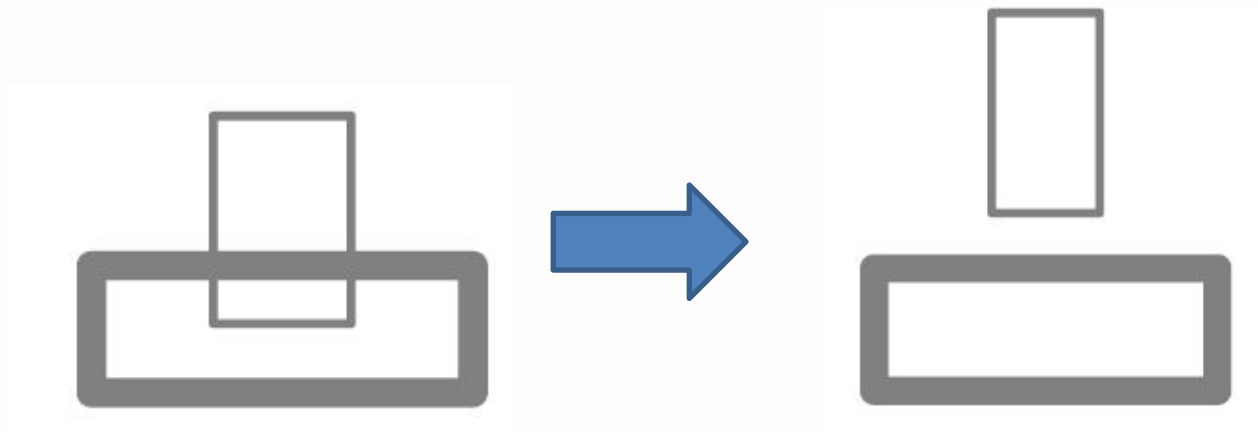
- Our eyes seek the most “natural” and “smooth” path between objects, even though they may not exist

How would you separate these items? (1 minute)



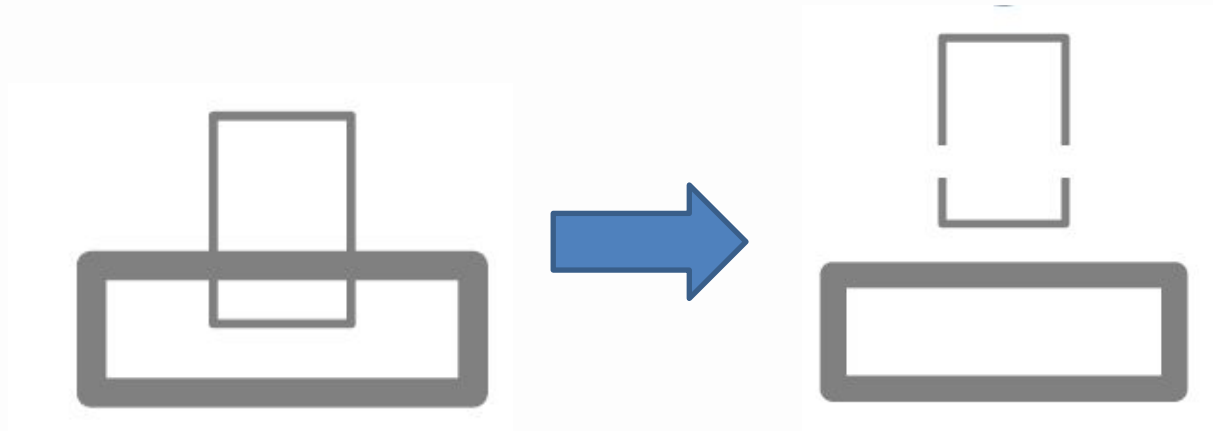
# Continuity (Continuidade) – 1

- Our eyes seek the most “natural” and “smooth” path between objects, even though they may not exist



## Continuity (Continuidade) – 2

- Our eyes seek the most “natural” and “smooth” path between objects, even though they may not exist



# Connection (Conexão)

- We tend to perceive connected objects as a group



- **Note:** connection is often stronger than color, shape or size

Analyze the following visualization (1-2 minutes).  
Next, you will be asked about it  
and the gestalt principles.

## Market size over time



# Which gestalt principles have been used?

- Proximity:
  - Indicates that the y axis, title and labels must be read together
  - Clarifies that the data labels and markers are related
- Similarity:
  - The similarity of colors (orange and blue) with the text is used to connect things

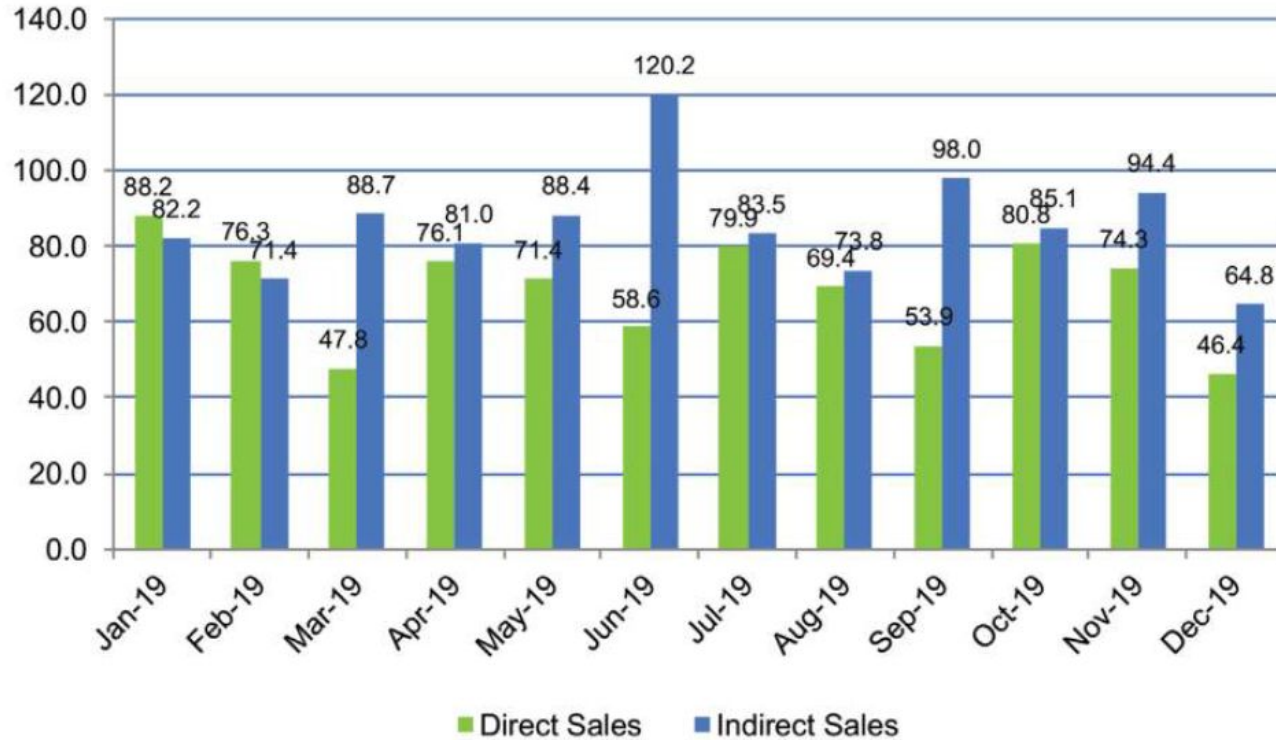
# Which gestalt principles have been used?

- Enclosure:
  - The gray region is used to differentiate the forecasts from the historical values
- Continuity:
  - The dashed line is used to connect the forecasts in the right section of the plot
- Connection:
  - In the line plot, all points are connected and make the trend easily visible

Which visual components would you change in the following visualization? (1-2 minutes)

## Time to Close Deal

Goal = 90 days



# 1. Removing the external blue lines

- The lines between the title and the plot, as well as the most external line are unnecessary
- The enclosure principle allows us to visualize the plot without them

## Time to Close Deal

Goal = 90 days



## **2. Removing the grid lines**

- Removing the grid lines, our attention is drawn to the data

## Time to Close Deal

Goal = 90 days

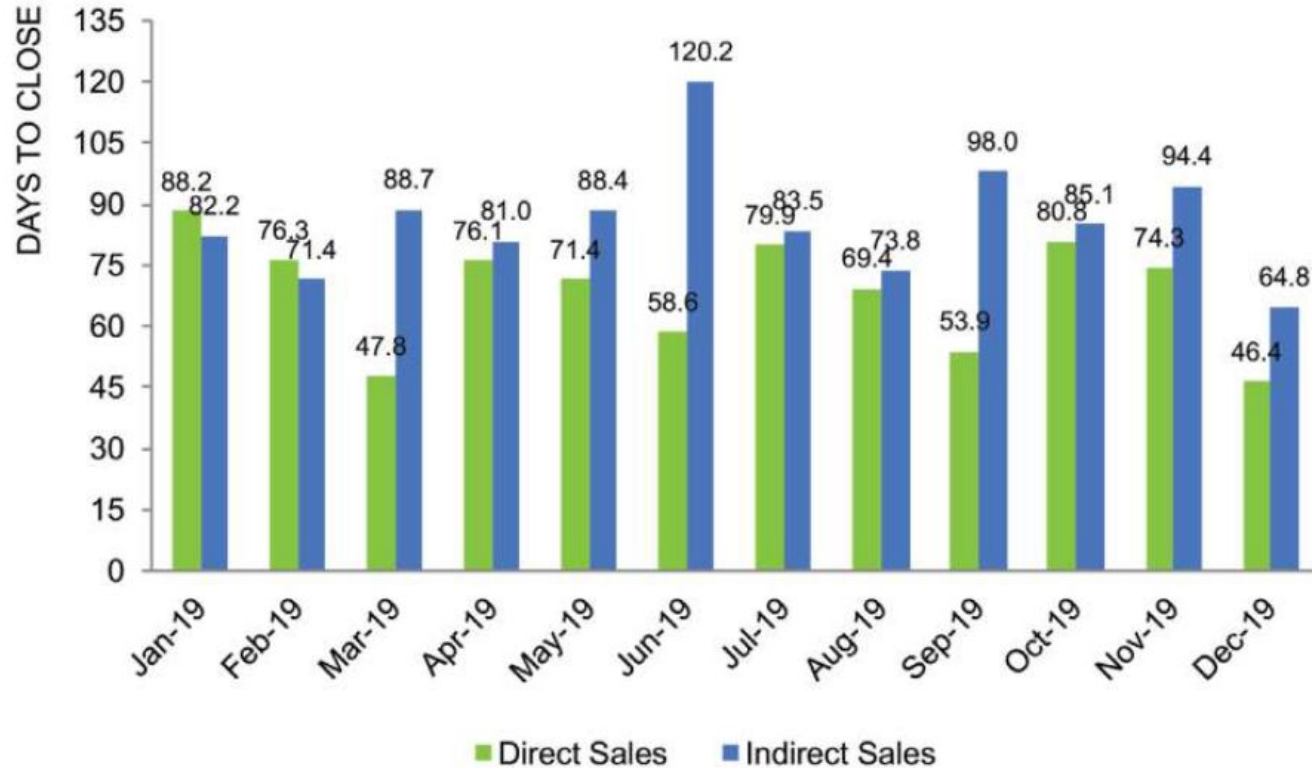


### **3. Improve the y axis**

- Remove the zeros from the y axis
  - The extra zeros in the decimal places are not required
- It is also interesting to change the y axis scale for 15-day intervals

## Time to Close Deal

Goal = 90 days



## 4. Improve the x axis

- Eliminate diagonal texts in the x axis
  - Diagonal and vertical texts are controversial
  - Whenever possible, prefer horizontal texts

## Time to Close Deal

Goal = 90 days

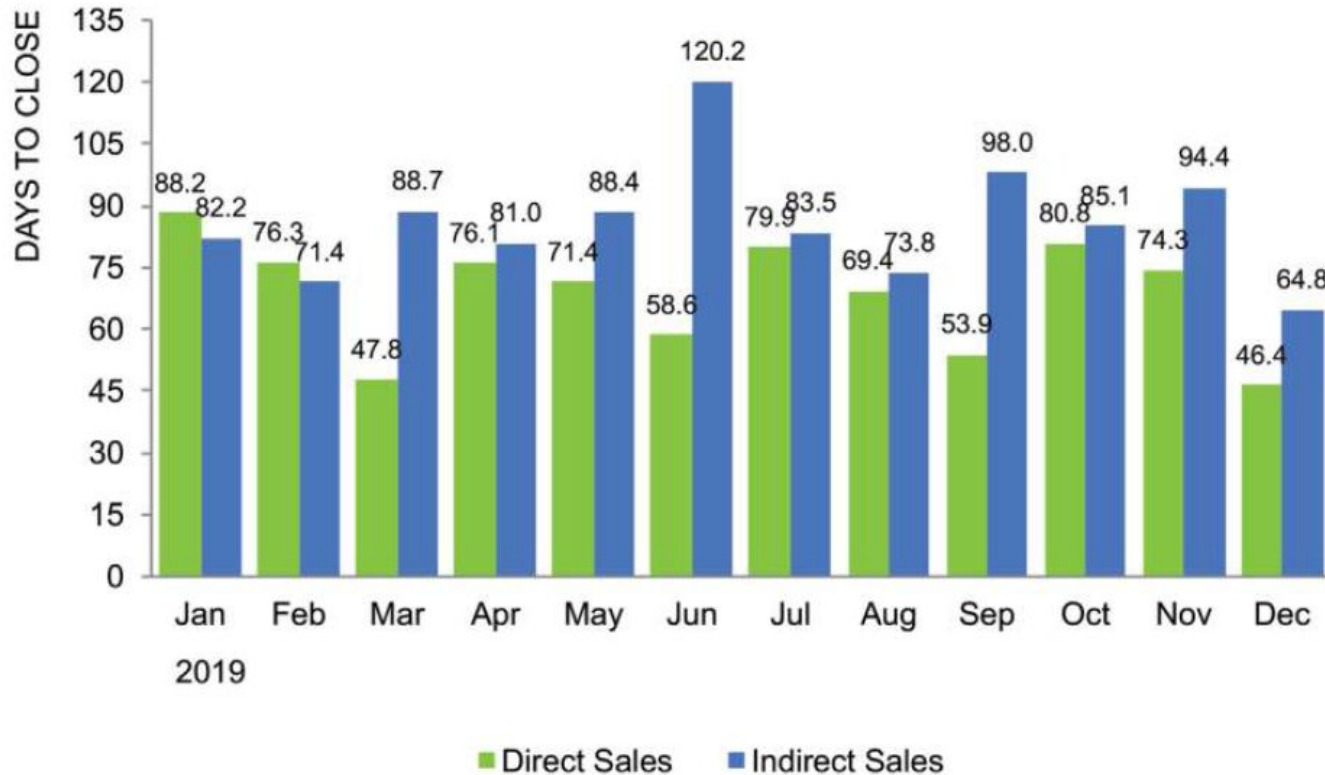


## **5. Decrease blank spaces**

- Avoid having unnecessarily big blank spaces between bars
- Useful due to the connection principle
- A good practice, however, is to keep blank spaces between bars from different categories

## Time to Close Deal

Goal = 90 days



## 6. “Drag” the labels to the bars

- Whenever possible, round the values

## Time to Close Deal

Goal = 90 days

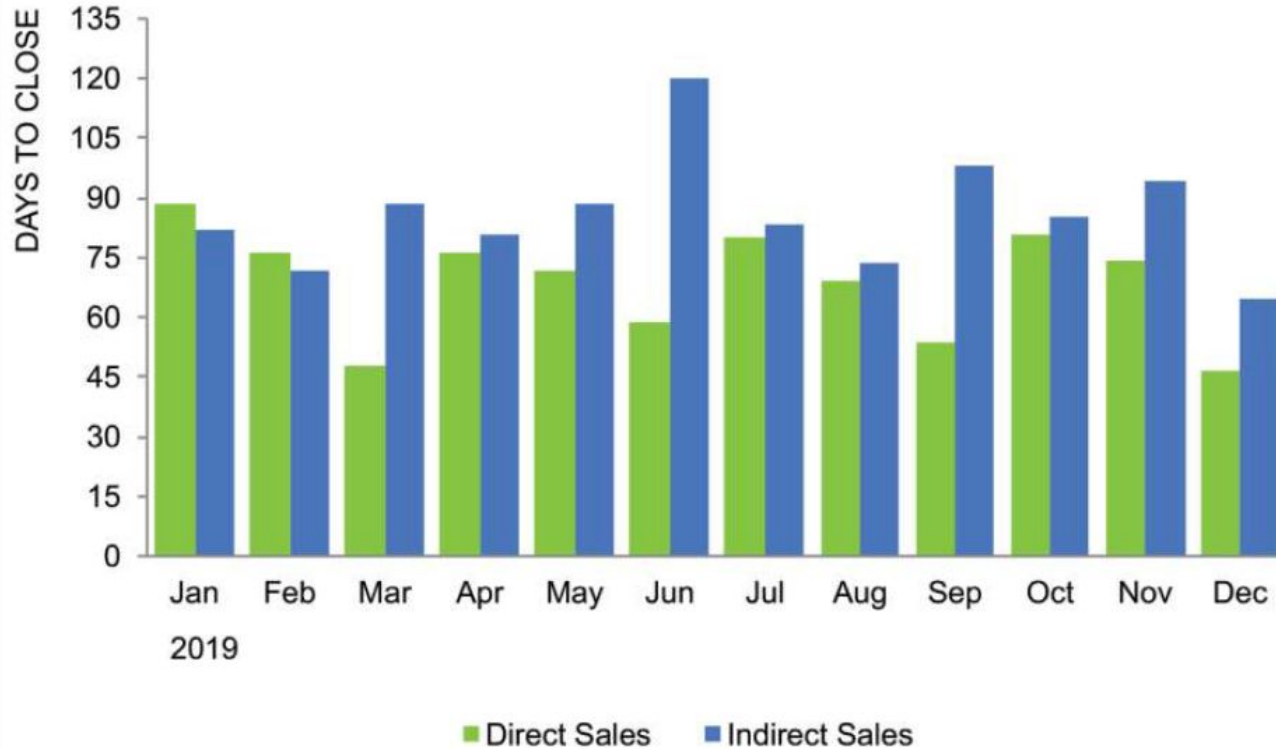


## 7. Eliminate the data labels

- The y axis is redundant with the numbers provided in the labels
- Important: remove or not to remove?
  - It depends on the context:
    - The exact values are required?
    - Or the trend is more relevant?

## *Time to Close Deal*

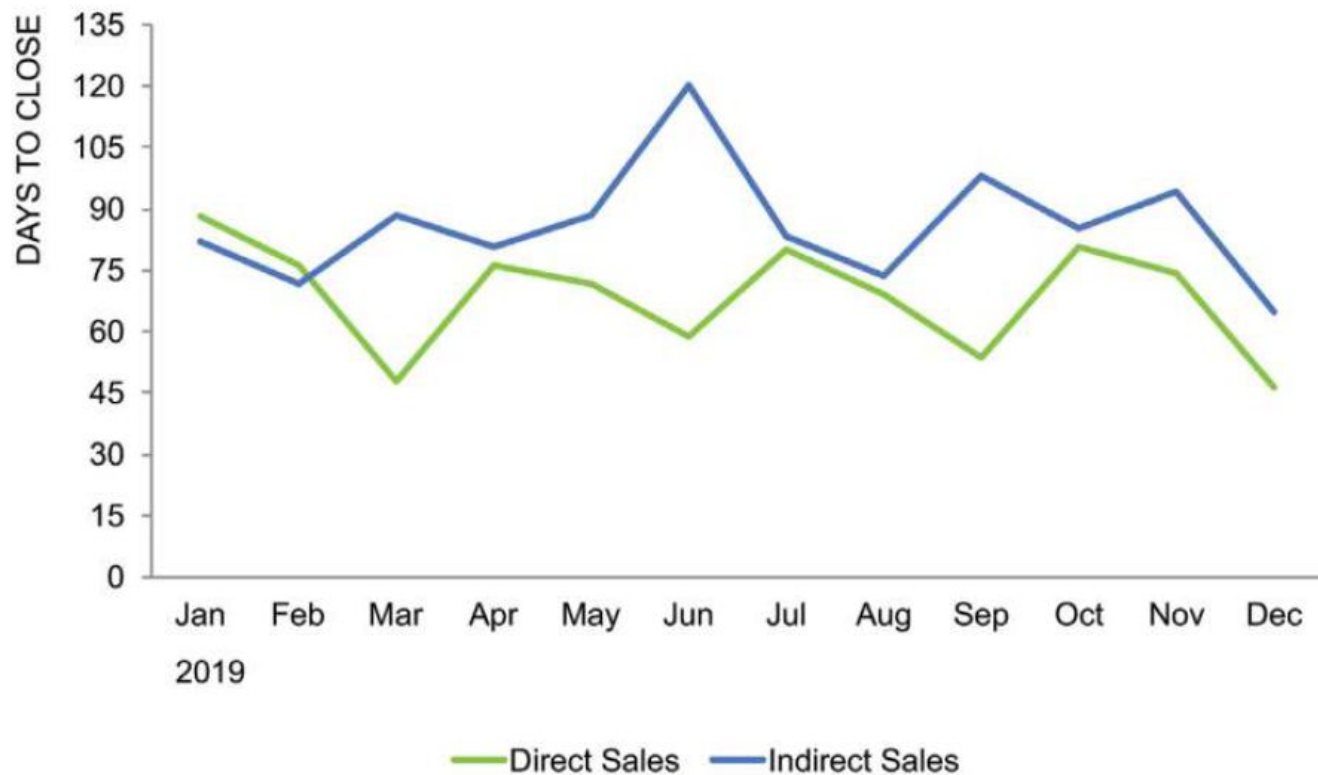
*Goal = 90 days*



## **8. Make it a line plot**

## Time to Close Deal

Goal = 90 days

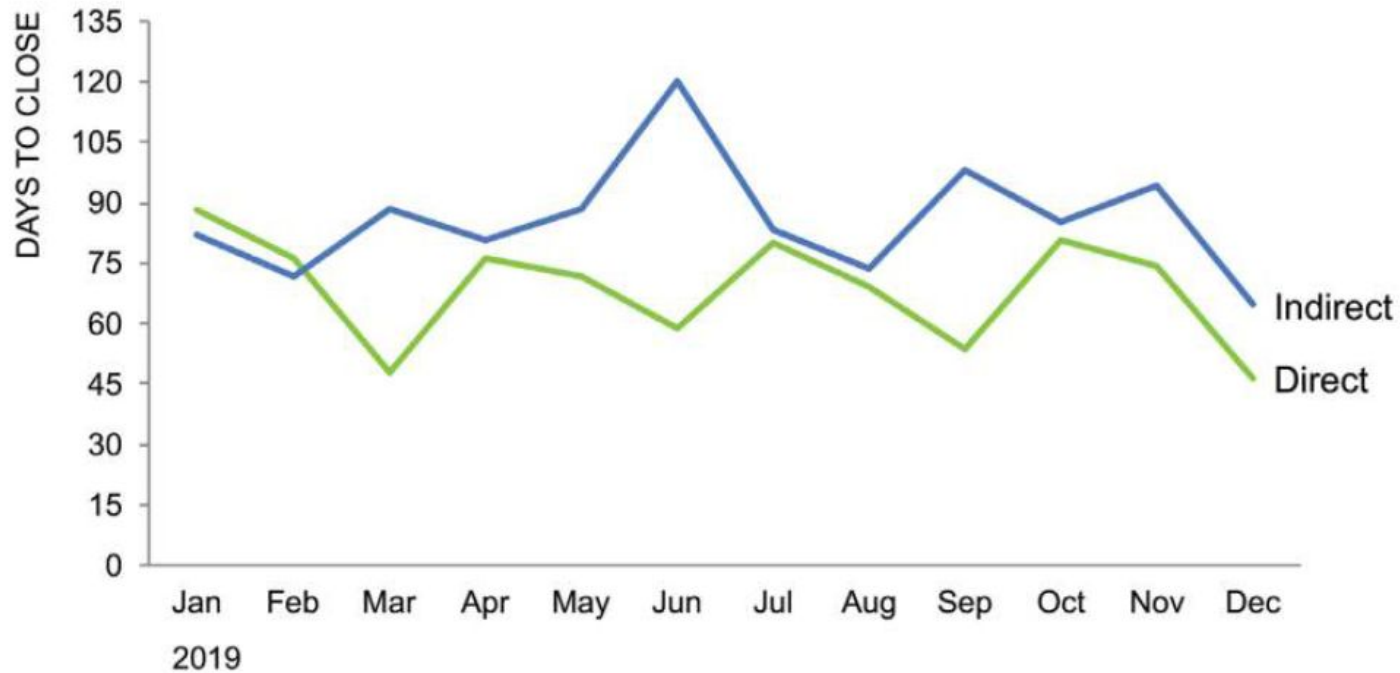


## **9. Apply the legend to the plot**

- Using the proximity principle, we can label our lines inside the plot itself

## *Time to Close Deal*

*Goal = 90 days*

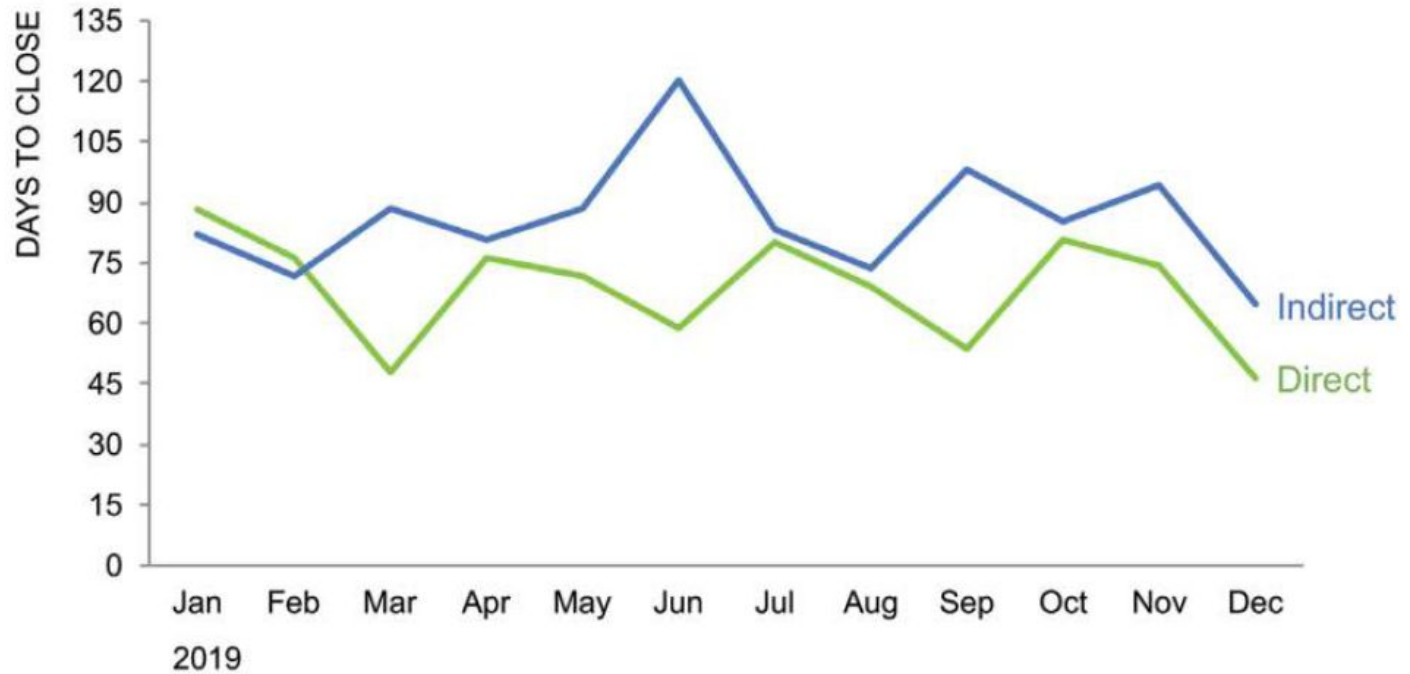


# 10. Changing the legend color

- Proximity and similarity

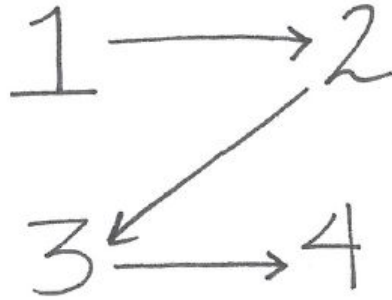
## *Time to Close Deal*

Goal = 90 days



# 11. Title Position

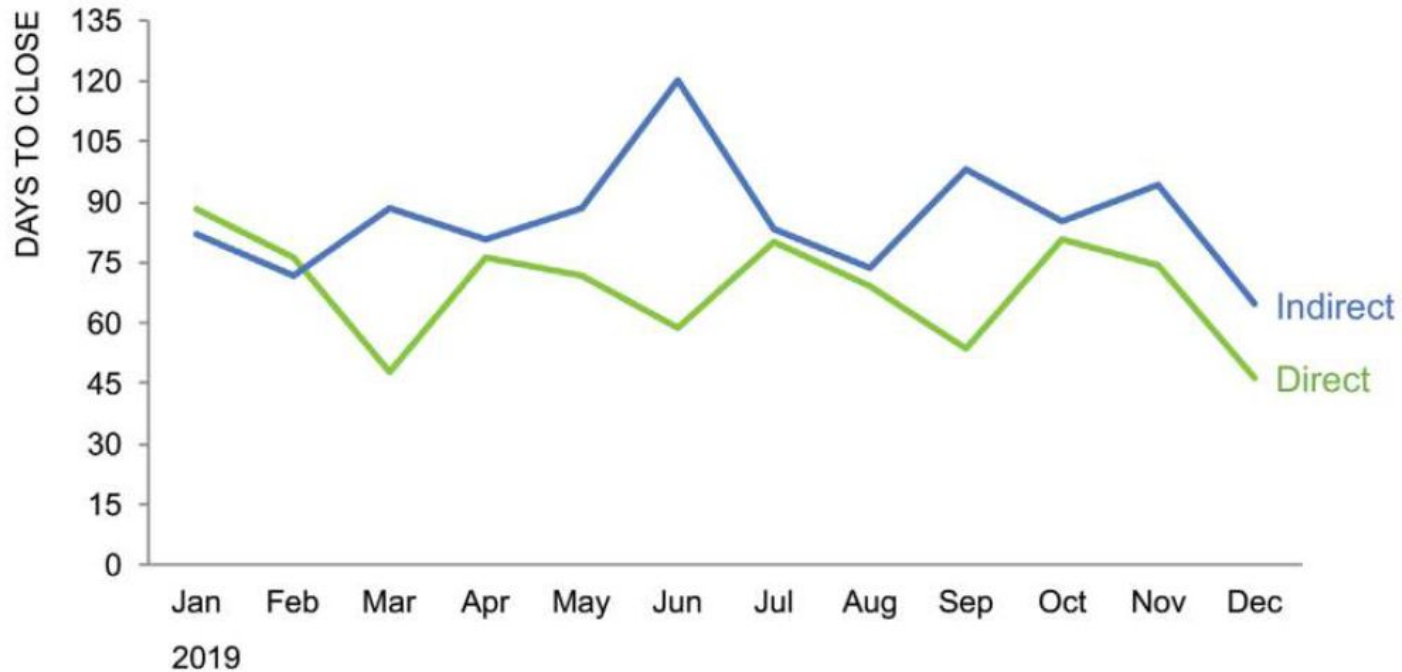
- Do not forget how we read (zigzagging z's):



- With this small change, the reader will focus on the title before anything else

## Time to Close Deal

Goal = 90 days

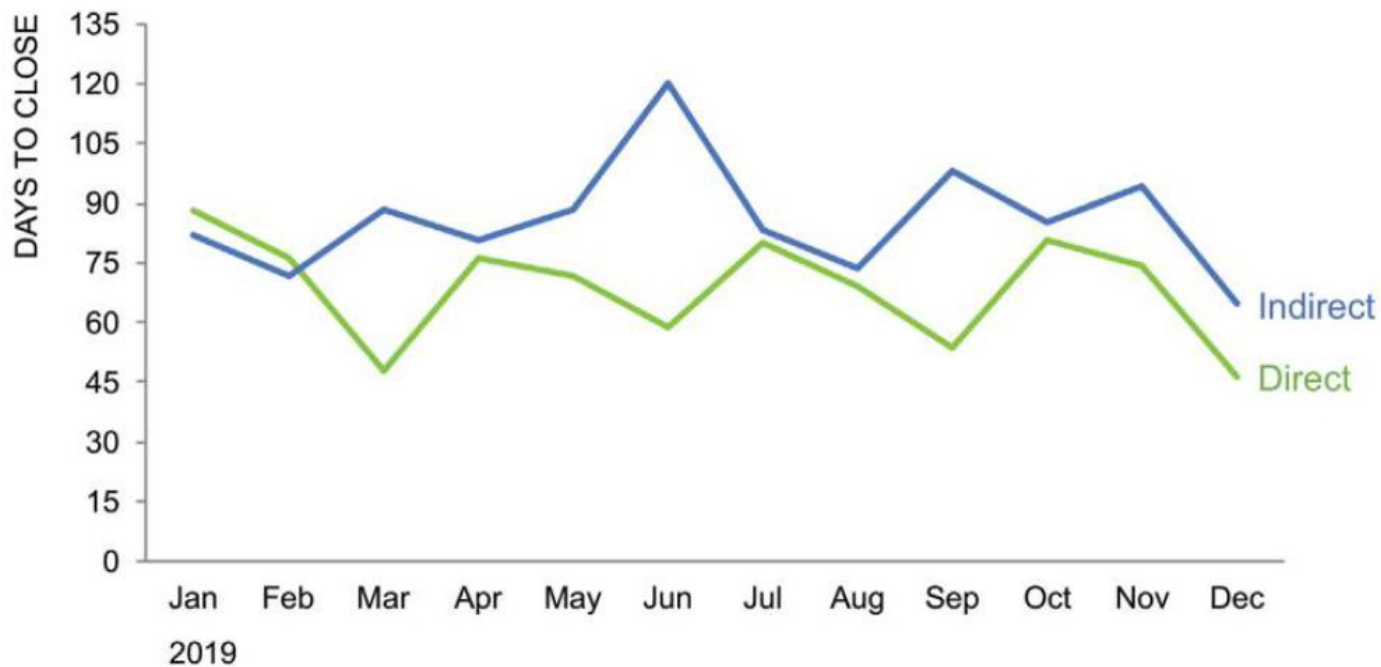


## 12. Removing the title color

- Is the color from the title anyhow related to the “indirect” component?

## Time to close deal

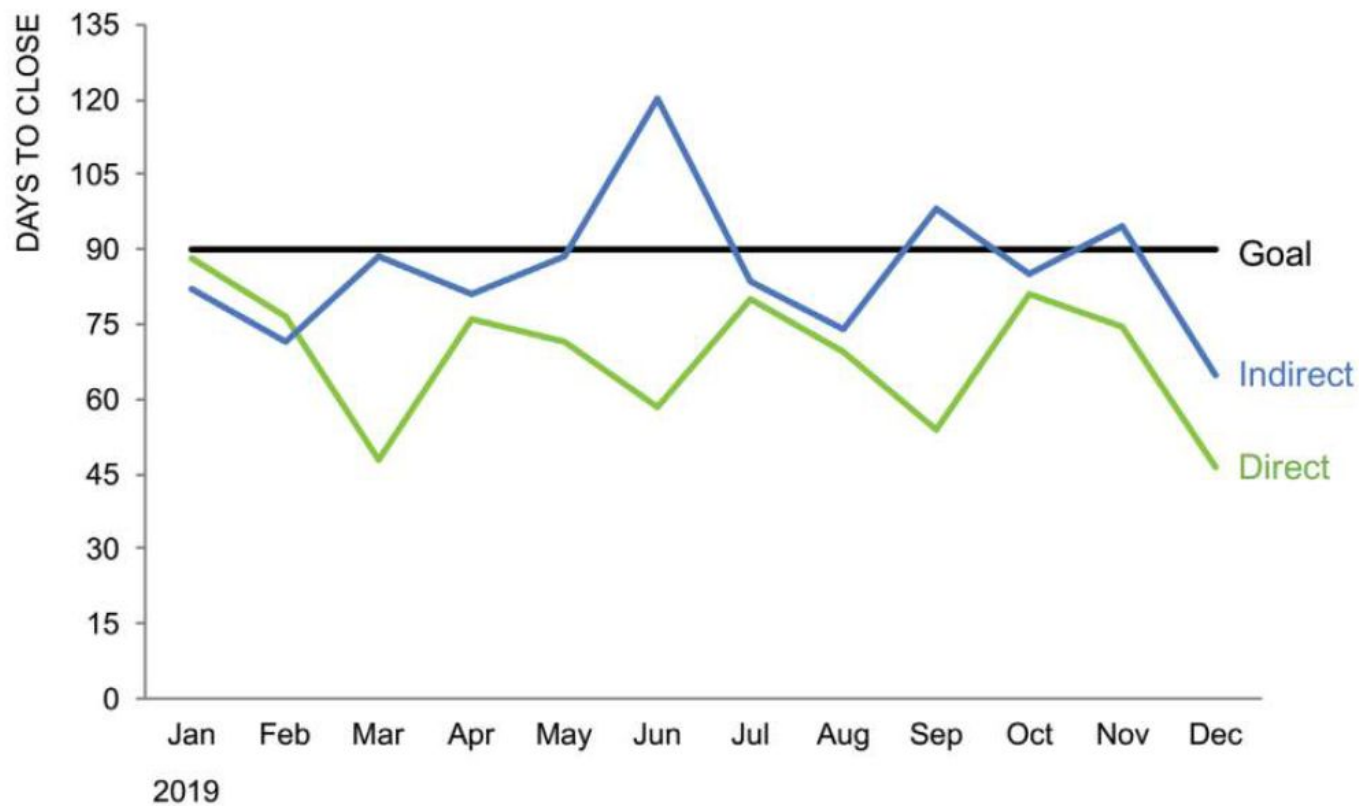
Goal = 90 days



## **13. Adding the Goal to the Plot**

- The goal to make a deal is 90 days
- This information can be added to the plot to make the analysis of the curves more visual

## Time to close deal

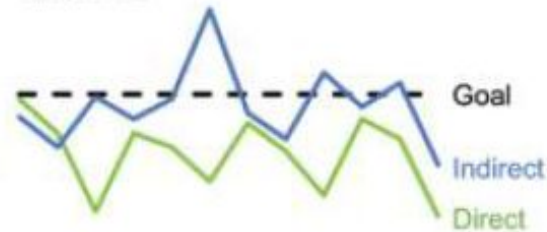


# 14. Testing different approaches

SOLID



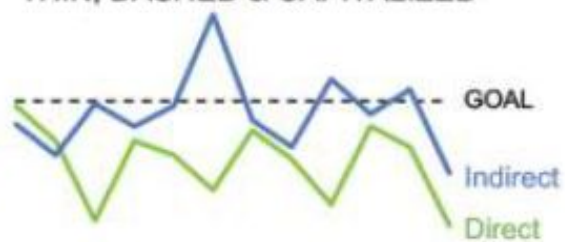
DASHED



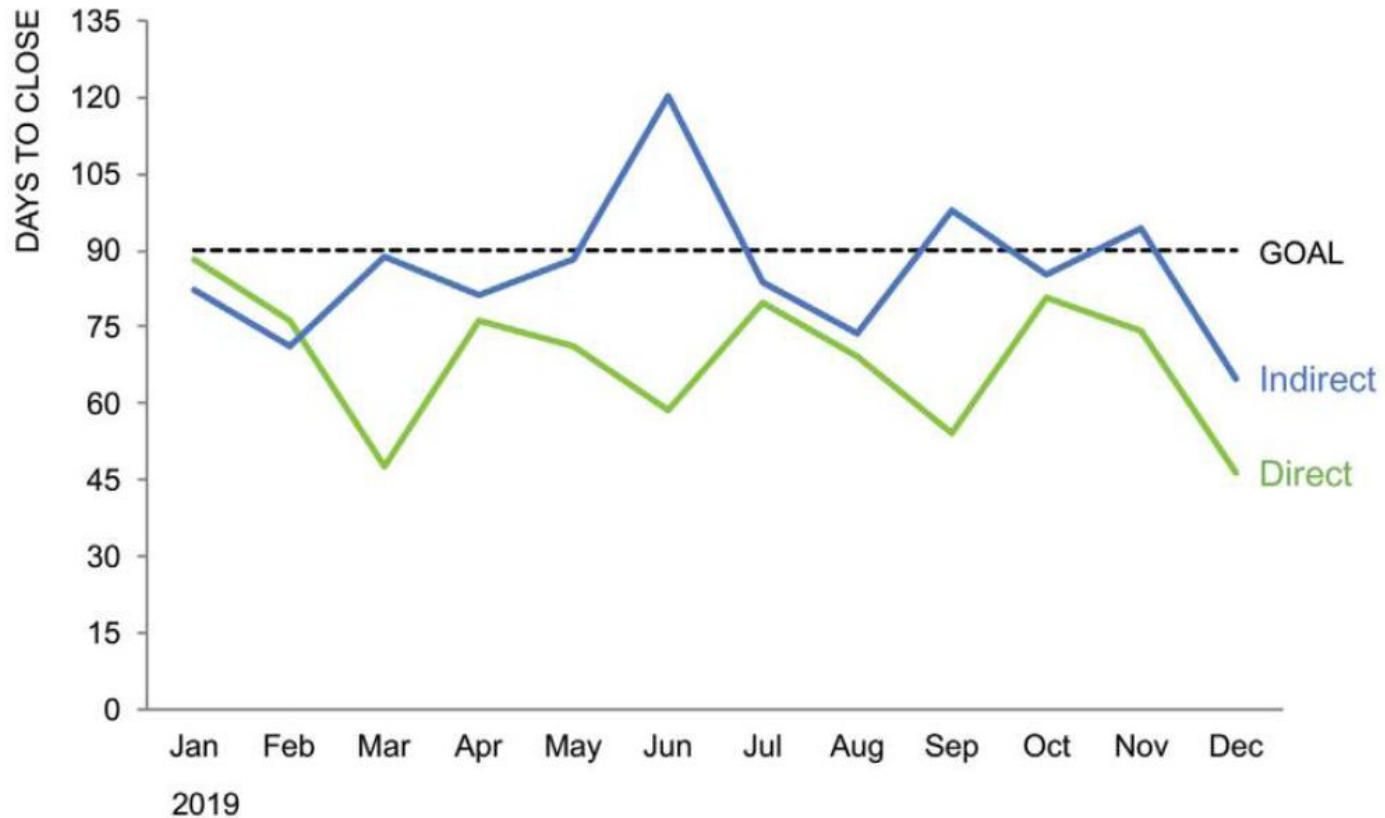
THIN & DASHED



THIN, DASHED & CAPITALIZED



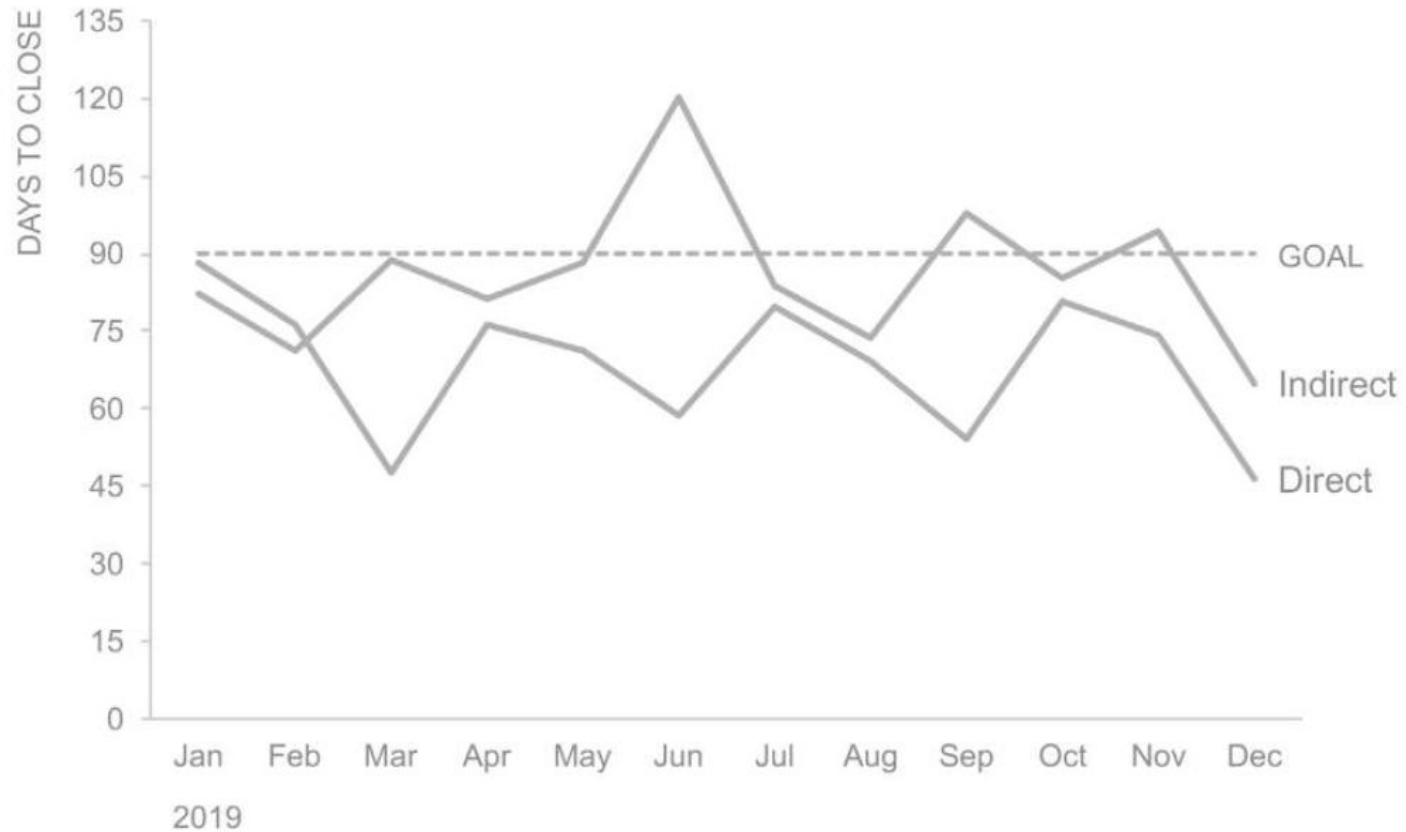
## Time to close deal



# 15. Removing Colors

- We have enough separation between the lines

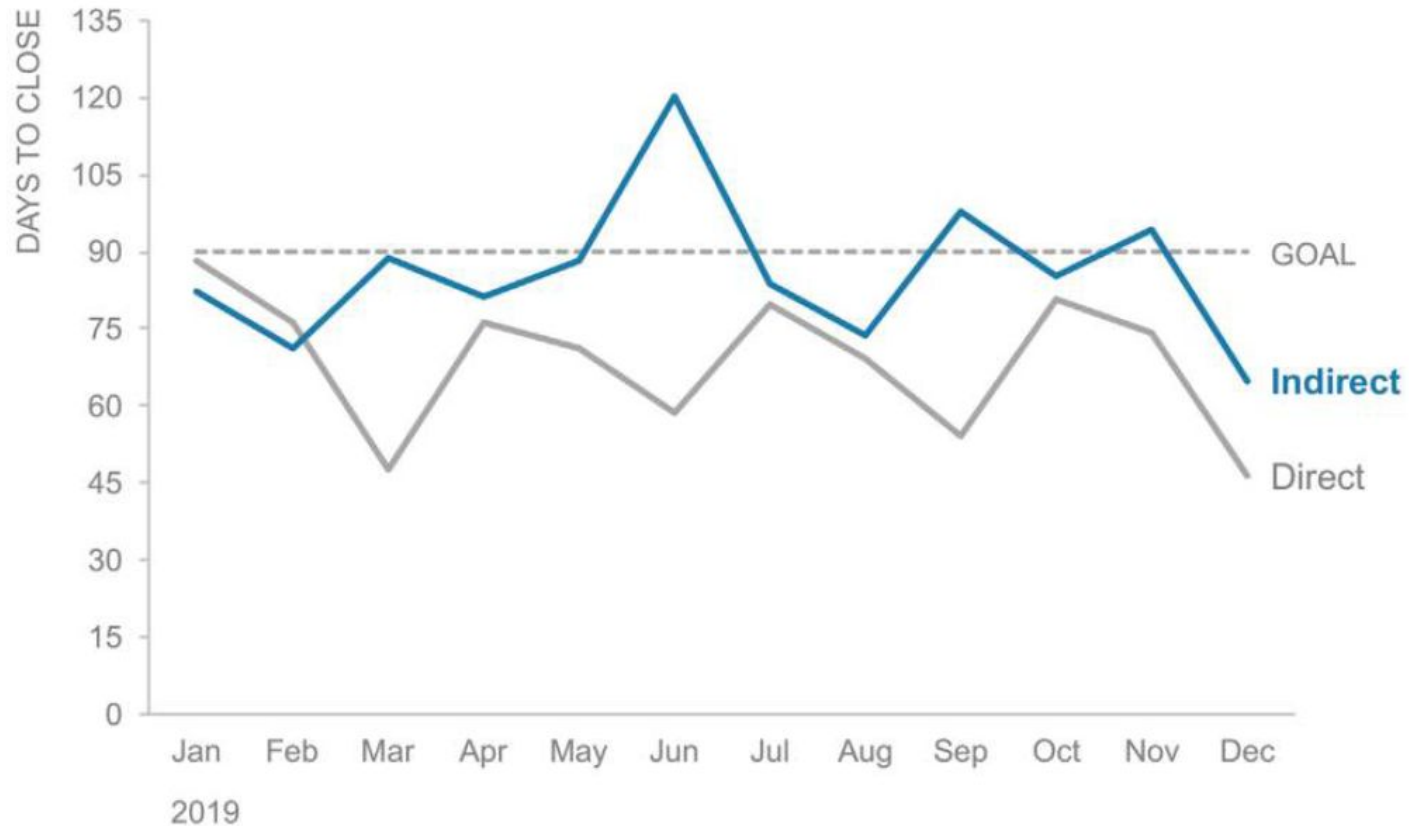
## Time to close deal



## 16. Drawing Attention

- Depending on the audience and goal of the visualization, we may draw the attention to one of the lines

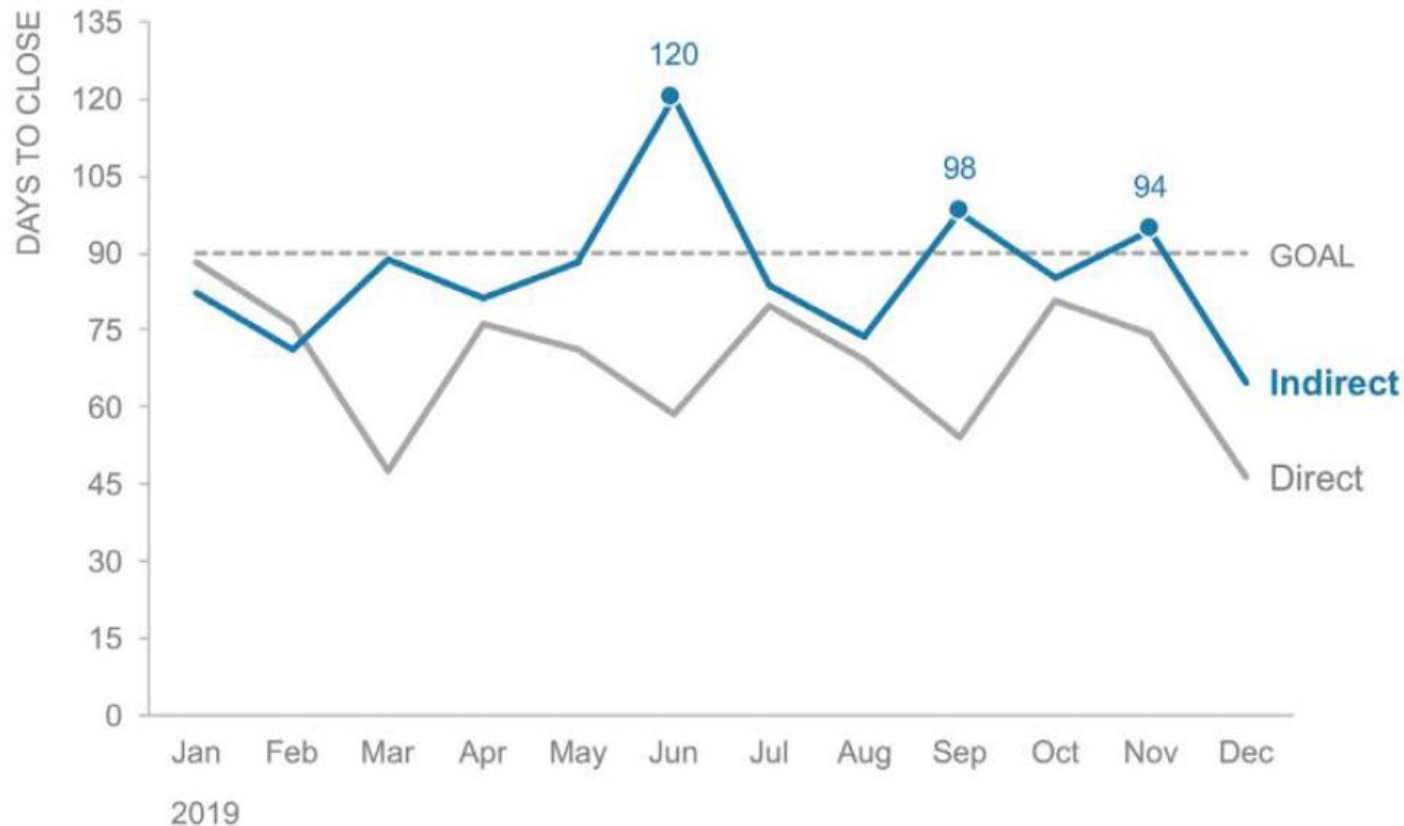
## Time to close deal: **indirect** varies over time



## **17. Focusing on Other Aspects**

- We can focus on other aspects, depending on what we intend to highlight

## Time to close deal: **indirect sales missed goal 3 times**

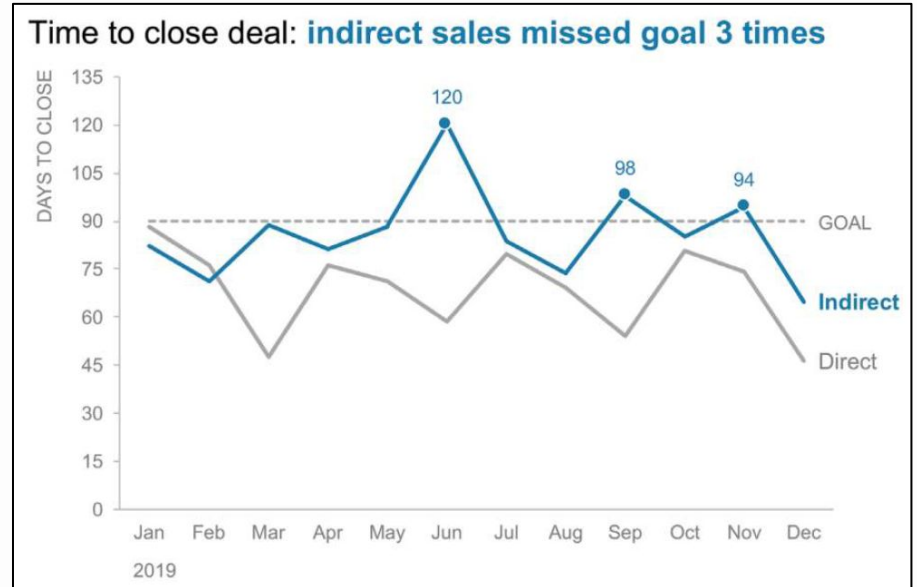


## Time to Close Deal

Goal = 90 days



# Initial vs. Final



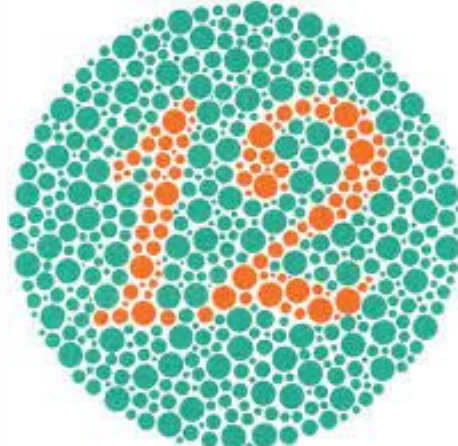
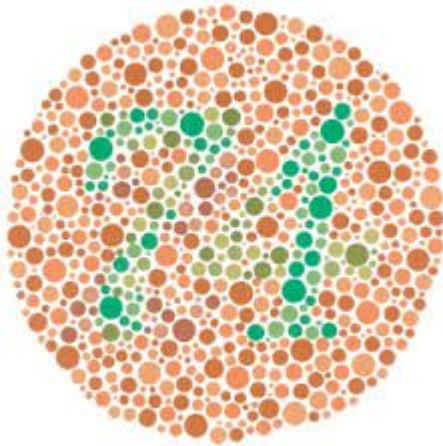
**BE MINDFUL OF COLORS**

# Colors

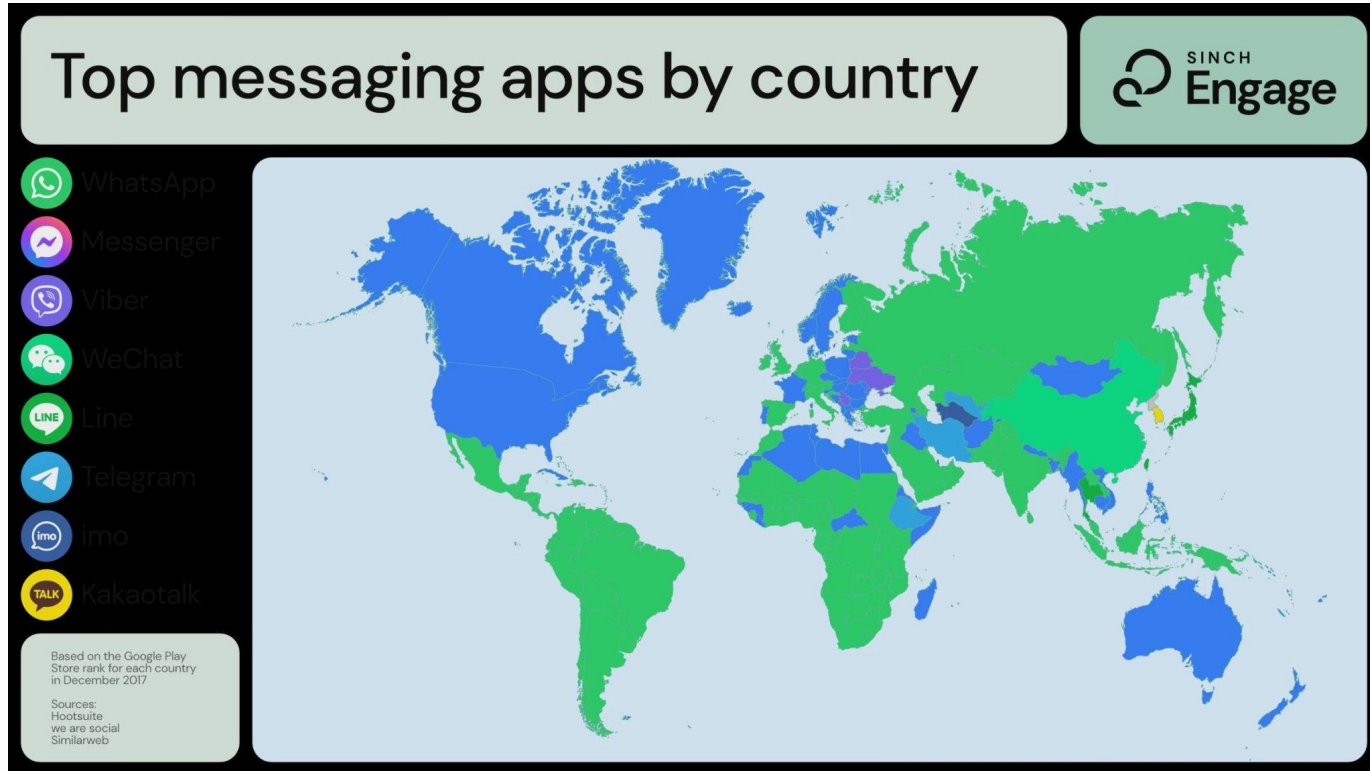
- One of the most common mistakes in visualizations regards the poor selection of colors
- Generally, all visualizations should use 2 colors, unless more are indeed needed
- Colors can be used to highlight things
- If colors are needed, avoid intense colors
  - Prefer colors with higher gray values

# Color Blindness

- Keep in mind: approximately 1 in every 12 men and 1 in every 200 women are colorblind!



# Color Blindness



# Project

- Use the remainder of today's class to work on your project.
  - You should be able to criticize your own visualizations, specially those in the Final Plots (**Effective Data Visualization**) section